

A case study on the integration of a snapshot hyperspectral field-portable imager solving fruit quality assessment

SeongHyun Cho^a, Eli Sheppard^a, Elvira Castello^a, Alexander Spanellis^a, Daniel Pearce^a, and Steve Chappell^a

^aLiving Optics LTD, 100 Park Drive, Milton Park, Abingdon, OX14 4RY, United Kingdom

ABSTRACT

This case study investigates the integration of a snapshot hyperspectral, field-portable imaging system for comprehensive fruit quality and yield assessment in agricultural orchards. The study introduces a novel framework for segmenting, counting, sizing, and estimating the ripeness of orchard fruit using spatial-spectral data. By combining hyperspectral imaging (HSI) with RGB data, we address the complexities of real-world agricultural environments, including occlusion, shadowing, and motion blur.

Our proposed 1D-ConvNet + SAM2 model outperforms the RGB-only YOLOv8s approach, achieving a 25 percent improvement in recall by leveraging spectral data to detect partially occluded fruit. Using DepthAnythingV2 (DAv2), we estimate metric depth to localize fruit detections within a unified 3D coordinate system, enabling unique tracking and accurate fruit counting across video frames. This capability supports yield estimation by providing precise counts and sizes of individual fruits in the orchard. Furthermore, segmentation masks and point clouds derived from monocular depth estimation are employed for fruit sizing, with future work focusing on improving handling of occlusions and segmentation errors.

In addition to spatial-spectral detection, we demonstrate the feasibility of non-destructive ripeness estimation by correlating spectral features in the fruit's skin with sugar content. An in-lab experiment using the Living Optics hyperspectral camera achieved a Brix correlation with an R^2 of 0.71, underscoring the potential of HSI for assessing ripeness and fruit quality.

This research highlights the utility of snapshot hyperspectral imaging as an all-in-one solution for end-to-end fruit quality and yield assessment in orchards. The dataset used in this study is publicly available at <https://huggingface.co/datasets/LivingOptics/hyperspectral-orchard>, encouraging further advancements in agricultural hyperspectral imaging and computer vision.

1. INTRODUCTION

Hyperspectral imaging (HSI) has emerged as a transformative technology for solving complex challenges in fruit quality and yield assessment, particularly in the context of precision agriculture. Unlike traditional imaging techniques that rely on the visible spectrum's three primary colors (red, green, and blue), HSI captures a wide range of wavelengths across the electromagnetic spectrum, including visible, infrared, and ultraviolet regions. This broad spectral coverage allows HSI to extract detailed spectral signatures, unique patterns that describe how materials reflect, absorb, or emit light at different wavelengths.^{1,2} By leveraging these spectral signatures, HSI can differentiate between materials and identify subtle biochemical and structural changes, even when objects appear visually identical.^{3,4}

HSI has shown significant promise in various fields, including material identification, medical diagnostics, cultural heritage preservation, and environmental monitoring.^{5,6} In agriculture, it offers unique advantages for tasks such as fruit detection, sizing, ripeness estimation, and yield prediction.⁷⁻⁹ However, these applications face challenges, including the high dimensionality of spectral data, limited labeled training datasets, and variability in spectral signatures caused by environmental factors such as occlusion, shadowing, and motion blur.^{10,11} Advancing the capabilities of HSI in real-world scenarios remains an active area of research, with a need for

Further author information: send correspondence to Alexander Spanellis e-mail: alex@livingoptics.com

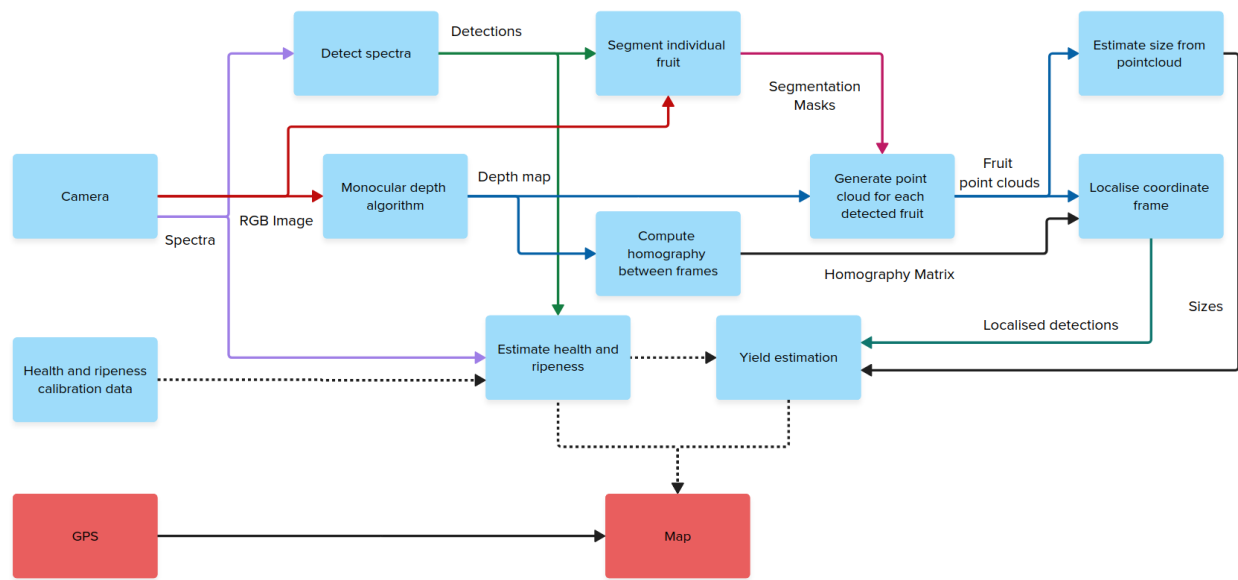


Figure 1: System overview for a hyperspectral detection, segmentation, tracking and quantification framework. Components in Blue are fully implemented, integrated and tested. Components in Red are beyond the scope of this work and their full development and integration will be the scope of future work.

robust frameworks that integrate HSI with complementary technologies like RGB imaging and advanced machine learning algorithms.

This paper presents a case study on integrating a snapshot hyperspectral, field-portable imaging system to solve fruit quality and yield assessment challenges in orchards. The study introduces a novel framework that combines spatial and spectral information to enhance segmentation, tracking, sizing, and ripeness estimation of orchard fruit in complex agricultural environments. Specifically, we explore the use of spectral data to complement RGB-based approaches, addressing limitations in traditional imaging systems. Our contributions are as follows:

1. **Establish Benchmarks for Spectral Sensing Technology:** We define and develop standardized benchmarks to assess the accuracy and effectiveness of spectral sensing technologies used in HSI for fruit data analysis. This provides a clear framework for evaluating the performance of various HSI systems in tasks such as detecting, classifying, and segmenting fruit.
2. **Highlight the Benefits of Spectral Sensing:** We explore the advantages of using spectral sensing technologies in the context of fruit data analysis, improving the precision of fruit detection and classification, enhancing the ability to track fruit growth, and providing valuable insights for agricultural applications.
3. **Identify Efficient Feature Recognition Methods:** We investigate efficient and effective methods for feature recognition by comparing the performance of spectral-based methods with spatial-spectral-based techniques. This demonstrates the potential of combining spectral and spatial information for improved detection and classification of fruit in diverse environments.
4. **Conduct Empirical Experiments:** We perform a series of empirical experiments to compare the performance of algorithms used in fruit data analysis, including tasks such as counting, tracking, sizing, and ripeness estimation. These experiments utilize the same data and evaluation framework to ensure fairness and consistency in evaluating different methods.

5. **Address Real-World Challenges:** We identify practical challenges, including fruit occlusion, varying lighting conditions, and environmental factors, that affect the accuracy of detection and classification systems. Solutions based on shared results from machine learning approaches applied to fruit data are proposed.
6. **Evaluate Performance Metrics:** We introduce and evaluate performance metrics that accurately measure the capabilities of HSI systems, providing robust indicators for their practical applicability in real-world scenarios.

Figure 1 shows an overview of the fruit quality assessment system that we have implemented for orchard fruit yield estimation. This system will be described and evaluated in this work. The methods and findings discussed in this study are not only applicable to orchard fruit but can also be adapted to other domains where hyperspectral imaging is valuable. Furthermore, a public dataset is made available at <https://huggingface.co/datasets/LivingOptics/hyperspectral-orchard> to encourage further advancements in hyperspectral imaging and agricultural computer vision. By addressing real-world challenges and delivering actionable insights, this study aims to advance the practical application of hyperspectral imaging in precision agriculture.

2. RELATED WORKS

2.1 Evolution of HSI Classification Techniques

The evolution of HSI classification has seen significant advancements, transitioning from traditional machine learning techniques such as k-Nearest Neighbors (k-NN),¹² Support Vector Machines (SVM),¹³ and Random Forests (RF)¹⁴ to more sophisticated methods. While these early approaches addressed high-dimensional spectral data challenges, they were limited in handling the data's complexity and scale. In recent years, advanced techniques like multinomial logistic regression¹⁵ and deep learning models—such as transformers,¹⁶ stacked autoencoders (SAEs),¹⁷ and Convolutional Neural Networks (ConvNets)¹⁸—have gained prominence. These methods leverage spatial correlations between spectral pixels, significantly improving classification performance when applied effectively.¹⁹

2.2 Spectral and Spatial-spectral Approaches

HSI classification approaches can be broadly categorised into spectral-based and spatial-spectral-based methods. Spectral-based approaches focus solely on the spectral signatures of individual pixels, often overlooking spatial dependencies.²⁰ Although these methods capture detailed spectral information, they may lack the spatial context necessary for higher accuracy.²¹ Spatial-spectral approaches, in contrast, integrate both spectral and spatial features, yielding enhanced classification results. Among these, ConvNets have demonstrated exceptional performance due to their ability to effectively combine spatial and spectral information, especially for agricultural tasks such as fruit classification.^{22,23}

2.3 Deep Learning and Recent Advances in HSI Classification

Recent advancements in deep learning have revolutionised HSI classification. A notable framework combining spectral and spatial features with SAEs, Principal Component Analysis (PCA), and logistic regression¹⁰ achieved higher accuracy than traditional methods like SVM and PCA classifiers. This study highlighted the importance of hierarchical feature extraction for improved classification, noting, however, that excessive network depth could hinder performance. Reviews of deep learning techniques for HSI emphasised challenges related to limited labelled datasets and spectral differences from traditional optical images,²⁴ while underscoring the effectiveness of 2D and 3D convolutional networks. Li et al.²⁵ categorised deep learning approaches into spectral-feature, spatial-feature, and spatial-spectral-feature networks, advocating for methods like reinforcement learning and data augmentation to address data scarcity.

2.4 HSI in Fruit Quality Assessment

HSI has emerged as a powerful tool for assessing fruit quality, enabling non-destructive evaluation of physical-chemical properties, maturity stages, and decay detection. A comprehensive review identified challenges in image processing, data analysis, and parameter optimisation, suggesting the integration of spectral and image data for enhanced utility.²⁶ Studies have also explored HSI's potential for early decay detection in fruits like apples and citrus, reporting promising results but acknowledging barriers such as large data sizes, high hardware costs, and redundant data, which limit industrial adoption.²⁷

2.5 Dimensionality Reduction and Quality Assessment Techniques

Dimensionality reduction techniques, such as PCA and Linear Discriminant Analysis (LDA), have been extensively employed to simplify HSI data processing. Applications include distinguishing banana maturity stages under varying temperatures, where HSI effectively correlated spectral data with chemical transformations during ripening, such as the conversion of starch into soluble sugars like glucose and fructose.²⁸ HSI's capabilities in evaluating textural, biochemical, and safety attributes of fruits and vegetables have further highlighted its non-destructive nature and real-time application potential, despite challenges like computational demands and morphological calibration.²⁹

2.6 Innovations in Fruit Ripeness, Maturity and Health Estimation

The assessment of fruit ripeness and maturity using HSI has gained traction due to its precision and non-invasive nature. Studies have employed techniques like band selection and dimensionality reduction to classify fruit maturity stages with high accuracy. For instance, blueberry maturity stages were classified with accuracies exceeding 88% using methods such as pairwise class discriminability and hierarchical dimensionality reduction alongside classifiers like k-NN, SVM, and AdaBoost.³⁰ Similarly, persimmon ripeness was determined with 95.3% accuracy through feature wavelength analysis and texture-based classification.³¹

HSI has also been used to automate ripeness evaluation in apples, combining spectral data with texture features to detect surface lesions and other quality parameters.³² By capturing internal and external fruit characteristics, these studies illustrate HSI's potential to enhance ripeness assessment accuracy significantly, paving the way for its integration into smart agricultural systems.

2.7 Methods and Challenges in Fruit Counting

Accurate fruit counting is vital for yield estimation, and various traditional and advanced methods have been developed to address this need. For instance, mango yield estimation using line-scan HSI on an unmanned vehicle achieved determination coefficients of up to 0.83, outperforming RGB methods.³³ Automated systems for pineapple crown counting have achieved 94.4% accuracy using UAV-captured RGB images combined with machine learning classifiers like artificial neural networks.³⁴ Similarly, texture- and shape-based methods have successfully counted mangoes on tree canopies, achieving counts within 16% of actual values under controlled imaging conditions.³⁵

2.8 Advances in Machine Learning for Fruit Counting

Machine learning has advanced fruit counting methods significantly. Studies comparing ConvNet architectures for detecting avocados, lemons, and apples demonstrated that Faster R-CNN achieves superior accuracy, albeit with higher computational demands, compared to single-shot detectors.³⁶ ConvNets have also been used to count clustered apples, achieving 97% accuracy by addressing challenges like occlusions and varying lighting conditions.³⁷ A low-cost green fruit counting system for orange trees reported a detection error of only 5% under controlled conditions, showcasing the potential for accessible technology adoption.³⁸

2.9 HSI for Fruit Detection and Size Estimation

HSI's versatility extends to fruit detection and size estimation. Reviews of deep learning techniques for these applications have highlighted efforts to overcome challenges like canopy occlusion and lighting variability.³⁹ Studies assessing peaches' internal and external quality attributes achieved high accuracy in weight prediction and diameter estimation with minimal error margins.⁴⁰ By integrating spectral and spatial data, HSI-based methods for fruit size and count estimation have consistently outperformed manual approaches, demonstrating its practical value in agriculture.⁴¹

2.10 Challenges in Yield Estimation Using HSI

Despite its potential, HSI faces significant challenges in yield estimation, including the misclassification of young leaves and overlapping fruit due to spectral similarities and canopy occlusion.⁴² Environmental variables such as inconsistent lighting, shadows, and debris further complicate data acquisition in outdoor conditions. Computational demands for processing high-dimensional data and the scalability of HSI systems remain key hurdles, particularly in commercial agricultural applications.

The processing of high-dimensional hyperspectral data poses significant computational challenges. Traditional techniques, such as linear unmixing, have been used to estimate vegetation abundance and correlate it with yield by analyzing multivariate imagery. While these methods achieve reasonable accuracy, they struggle with nonlinear interactions and require extensive preprocessing to address errors caused by mixed spectral signatures in heterogeneous canopies.⁴³

Deep learning models, like Convolutional Neural Networks (CNNs), have shown promise for yield estimation by combining spectral and spatial information. For example, a CNN-based approach integrating spectral and RGB data achieved a prediction accuracy of 75.5% for corn yields, surpassing one-dimensional and two-dimensional models. However, the approach faced limitations, including small dataset sizes and a reliance on preprocessing to optimize performance, emphasizing the need for larger datasets and robust preprocessing pipelines for broader applicability.⁴⁴

In orchard environments, ground-based HSI has been employed to estimate mango yields with results comparable to RGB-based methods. Despite its advantages in detecting nuanced traits like disease and maturity, the cost-effectiveness of HSI as a standalone tool for yield estimation remains questionable. Its practical scalability is further hindered by labour-intensive ground truthing and the significant preprocessing demands of HSI data.³³

Efforts to integrate hyperspectral data with machine vision systems for automated fruit detection across growth stages have demonstrated potential. However, these systems face challenges in developing algorithms that can handle variability in fruit appearance, such as differences in colour and size, across developmental stages and environmental conditions. Furthermore, achieving the real-time processing speeds necessary for large-scale agricultural operations remains an ongoing hurdle.⁴⁵

3. OVERVIEW

The remainder of this paper is structured as follows:

- IV. **Dataset Preparation** – This section describes the collection and preparation of the dataset, detailing its structure and the preprocessing steps involved.
- V. **Fruit Detection and Segmentation Methodology** – Here, we outline the methods developed for fruit detection using hyperspectral imaging and compare the performance of various approaches.
- VI. **Yield Estimation Through Object Tracking** – This section presents our tracking algorithm for detecting fruit and demonstrates how it can be applied to estimate the total yield in a row of trees.
- VII. **Fruit Sizing** – A method for analyzing the size distribution of fruit within a row of trees is discussed, providing insights into size variability and uniformity.
- VIII. **Ripeness Estimation** – This section explores a non-destructive approach for estimating fruit ripeness using hyperspectral imaging, emphasizing its potential to assess internal and external quality attributes effectively.
- IX. **Conclusion** – A summary of the key contributions and findings of this research is provided.
- X. **Future Work** – Finally, we discuss potential areas for improvement and further exploration to enhance the utility and scalability of hyperspectral imaging workflows.

4. DATASET PREPARATION

The dataset was acquired using a Living Optics hyperspectral camera,⁴⁶ which captures detailed spectral data spanning the visible to near-infrared (VIS-NIR) range, covering wavelengths from 440nm to 900nm. Each data sample includes both standard RGB images with a resolution of $2048 \times 2432 \times 3$ and hyperspectral (HS) data, comprising 96 narrow spectral bands from 4384 spectral samples. For simplicity, we refer to the combined RGB and spectral data as hyperspectral (HS) images.

4.1 Data Scope and Collection

This dataset was developed in collaboration with the UK's largest orchard fruit producer and covers over 1,000 trees of multiple fruit varieties throughout the growing season (March to September). The dataset supports various applications, including fruit detection, sizing, segmentation, and disease detection, by offering a comprehensive view of fruit development stages.

4.2 Data Annotation and Splitting

A subset of the dataset has been annotated, encompassing 44 unique raw files and 439 frames. To ensure robust evaluation, an 8:2 train-test split is employed at the raw file level, so all frames from a given file are confined to either the training or testing set. This approach minimizes the risk of data leakage and maintains dataset integrity.

4.3 Class Distribution

The dataset includes three annotated fruit classes: Royal Gala Apple, Pear, and Cox Apple, with respective instance counts of 3,785, 2,523, and 73, respectively. This substantial class imbalance necessitates careful data preprocessing to address training challenges, particularly for the underrepresented Cox Apple class, which is present exclusively in the training set.

4.4 Background Class and Imbalance

Unlabelled portions of each scene are mined for spectral samples to create a background class, which adds substantial imbalance due to the high volume of background data compared to foreground spectra. Additional details are provided in [section 12](#).

4.5 Spectral Characteristics

The spectral profiles for each class, as shown in [Figure 2](#), demonstrate unique characteristics. Royal Gala Apples exhibit a pronounced peak between channels 42 and 62, associated with their strong red coloration. Pears exhibit spectral profiles similar to the background, primarily because of shared green hues. Lastly, the Cox Apple's profile indicates broad spectral responses across the visible range capturing changes akin to yellow-tone identifiers reflective of its pale appearance.

4.6 Labeling Spectra

Spectral samples are extracted from HS images and assigned labels based on the segmentation mask at their respective locations. This process produces two aligned arrays for each dataset split: a spectral array of shape $[N, 96]$, which represents the spectral samples, and a class array of shape $[N, 1]$, containing the corresponding class labels. Samples that fall outside labeled segmentation masks are assigned to the background class, which is labeled as 0, thereby contributing to the imbalance within the dataset.

4.7 Data Normalisation

To enhance the performance of spectral classification algorithms, two normalization steps are applied to the spectra:

Figure 2: Mean spectrum of each class in the dataset. The shaded area represents the standard deviation of each class.

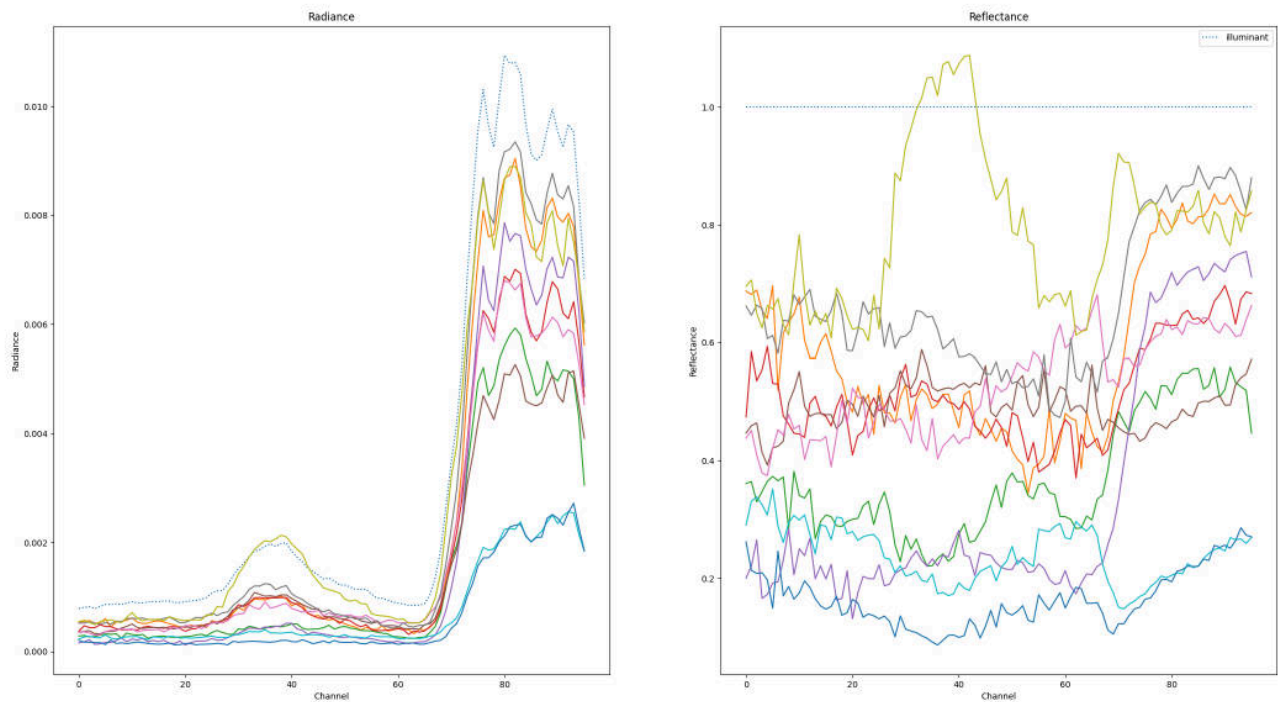
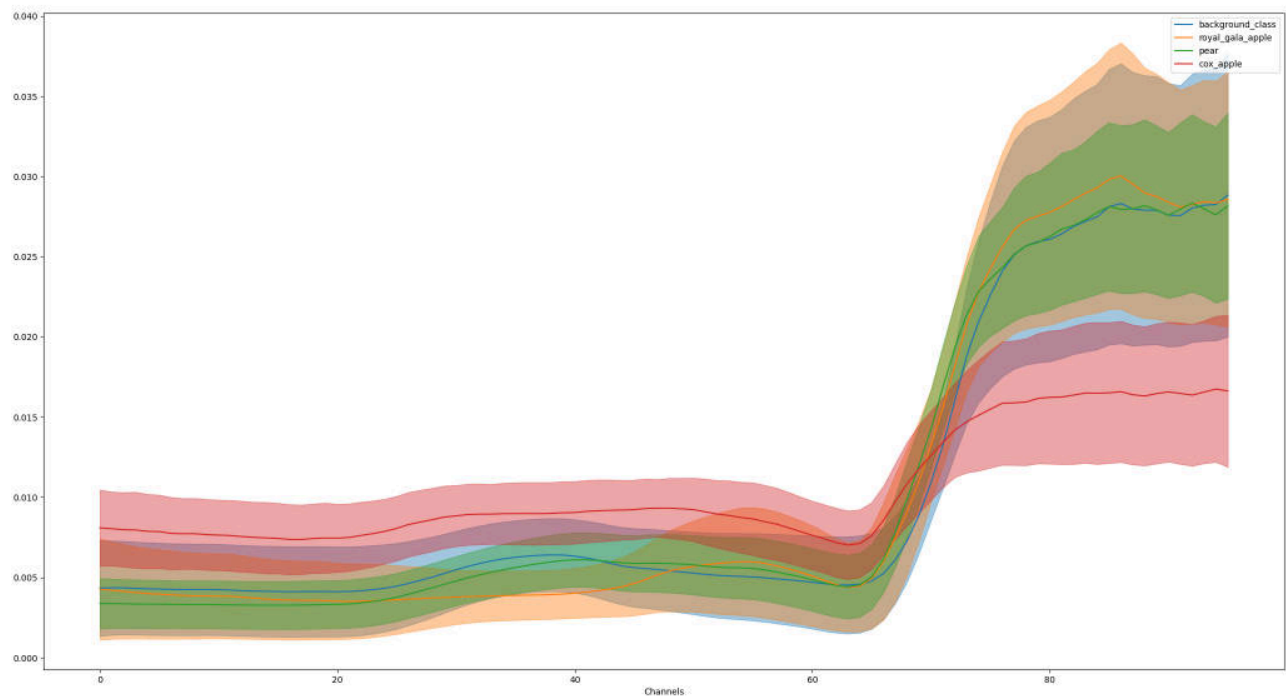


Figure 3: A selection of spectra and the estimated scene illuminant in radiance (left) and reflectance (right).

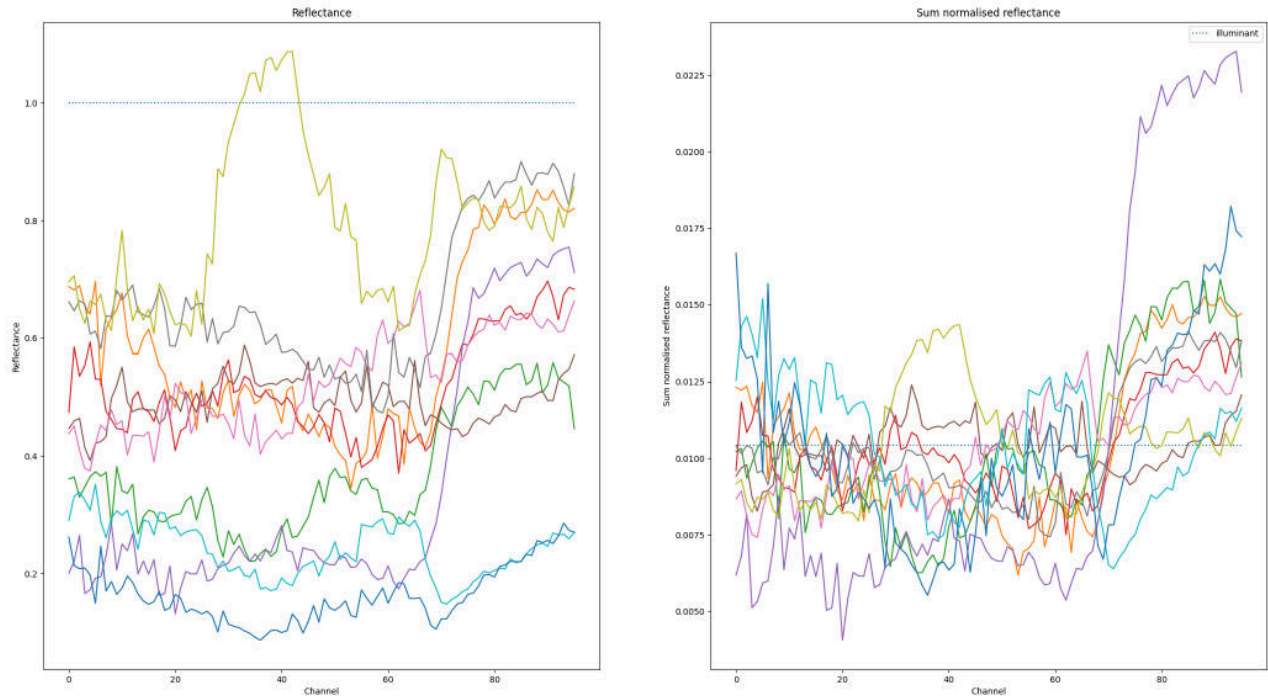


Figure 4: A selection of spectra and the estimated scene illuminant in reflectance (left) and sum normalised reflectance (right).

4.7.1 Reflectance Conversion

The average spectrum of the brightest 5% of pixels in each HS image is calculated to estimate the illuminant. Each spectrum is then divided by this illuminant to convert radiance to pseudo-reflectance, mitigating lighting variability between scenes. Figure 3 illustrates this process.

4.7.2 Sum Normalisation

Each spectrum is divided by its channel-wise sum to standardize the area under the curve to one. This step reduces intra-class variance caused by variations in camera settings (gain, exposure) and environmental factors like shadows. Examples are shown in Figure 4.

5. FRUIT DETECTION AND SEGMENTATION METHODOLOGY

5.1 Experimental Setup

The image dimensions vary by model type. The Spectral Segmentation Model leverages the full spectral resolution of 4384 spectral points, while the Spatial-Spectral and YOLOv8s models operate on standardized images resized to 960x960 pixels (cropped and half resolution). This resolution limitation arises due to the Living Optics Camera, which collects spectral samples only in the central 1920×1920 region due to its optical design.⁴⁶ As a result, only objects within this central area are labeled, leaving regions outside this area without ground truth segmentation masks.

This study benchmarks the performance of seven models on spectral detection tasks with HS and RGB data. These models, including two Spectral detection algorithms, four Spatial-Spectral segmentation methods, and one RGB segmentation model, employ diverse strategies to optimize detection accuracy and computational efficiency for HS and RGB analysis challenges.

5.2 Spectral Detection Algorithms

This section details two spectral detection models used for hyperspectral data classification: a RF classifier as a baseline and a 1D Convolutional Neural Network (1D-ConvNet) as a more advanced approach. Each model leverages spectral information to predict class labels for individual spectra.

5.2.1 Random Forest

The RF classifier is implemented using the scikit-learn library, employing the OneVsRestClassifier framework for multiclass classification. It predicts the most probable class for each spectrum. Custom hyperparameters, such as a minimum of two samples per leaf, five samples per split, and 400 estimators, are used to configure the model. Class weighting strategies are also applied to address the dataset's imbalance, as discussed in [section 12](#).

While the RF model achieves perfect accuracy on training data, it tends to overfit due to its decision trees extending fully until leaf nodes are pure. Attempts to mitigate overfitting by limiting the maximum depth of trees result in decreased test set performance. This outcome suggests that preprocessing steps, such as reflectance conversion and sum normalization ([subsection 4.7](#)), have effectively produced well-separated classes, thereby reducing the need for additional regularization.

5.2.2 1D Convolutional Classifier

The second method employs a 1D-ConvNet for spectral classification. A grid search is conducted to refine the model's architecture and hyperparameters, starting with a foundational structure. This baseline consists of a single convolutional layer, a ReLU activation, batch normalization, a dropout layer to prevent overfitting, and a fully connected output layer with four neurons. To optimize performance, various configurations of convolutional and fully connected layers are tested during the grid search.

The final architecture identified through this process includes a single convolutional layer with 64 neurons, followed by four fully connected layers that progressively narrow from 512 to 64 neurons. The output layer applies Sigmoid and Softmax activations to compute class probabilities. Training is performed using the Adam optimizer with early stopping based on validation F1 scores, and the model achieves its best results by balancing robust spectral features with effective generalization.

The grid search explores two primary parameters to identify the optimal 1D-ConvNet architecture. The first involves varying the number and size of convolutional layers, ranging from one to four layers, with neuron counts set at 64, 128, 256, and 512. The second parameter involves the number and size of fully connected layers, tested across configurations ranging from zero to four layers beyond the output layer. Both increasing and decreasing layer-width patterns, such as [64, 128, 256, 512] and [512, 256, 128, 64], are considered. This combination results in a total of 32 potential architectures for evaluation.

The grid search is conducted using the "Background subsampling 1" method to address class imbalance. Training is performed with a batch size of 4384, equivalent to one frame of spectral data, and optimized using the Adam algorithm⁴⁷ with default hyperparameters. An early stopping mechanism, triggered after 100 epochs without improvement in the validation F1 score, is employed to prevent unnecessary computation. The final architecture selected is the one achieving the highest validation F1 score. As is common in architecture searches,⁴⁸ it is assumed that strong interim performance correlates with robust final results.

The selected architecture begins with a single convolutional layer comprising 64 neurons operating along the spectral channels. This is followed by a ReLU activation and batch normalization to stabilize the feature distributions. Beyond the convolutional block, the network contains four fully connected layers with progressively narrowing sizes of 512, 256, 128, and 64 neurons. The output layer includes four neurons, utilizing Sigmoid and Softmax activation functions to compute class probabilities.

A significant challenge in training the 1D-ConvNet arises from the pronounced imbalance in the orchard dataset. This imbalance manifests in two ways: the overwhelming prevalence of background spectra compared to foreground spectra and the uneven distribution of samples among the foreground classes. To counter these issues, various techniques are employed to ensure that the model effectively learns to distinguish all classes, despite the dataset's inherent disparities.

Table 1: Results of different data balancing methods on the optimised 1D convolutional classifier architecture.

Balancing scheme	Validation Accuracy				
	Background	Pear	Royal Gala Apple	Cox Apple	Total
None	1.0	0.0	0.0	N/A	0.97$\frac{1}{2}$
Class weighting	1.0	0.0	0.0	N/A	0.97$\frac{1}{2}$
Background subsampling 1	0.89	0.79	0.60	N/A	0.88
Background subsampling 2	0.16	0.59	0.54	N/A	0.18
Data duplication 1	0.99	0.20	0.06	N/A	0.97$\frac{1}{2}$
Data duplication 2	0.96	0.57	0.23	N/A	0.94

1. **None:** The dataset is used without any adjustments to address the imbalance, relying on the raw distribution of the data.
2. **Class Weighting:** Loss values are weighted inversely to the frequency of each class, giving more significance to rare classes while reducing the impact of common ones. This approach penalizes misclassification of rare classes more heavily, while frequent classes, due to their prevalence, contribute equivalently to the overall loss.
3. **Background Subsampling 1:** At each training epoch, a subset of background spectra is randomly sampled to match the total number of foreground spectra across all foreground classes.
4. **Background Subsampling 2:** A subset of background spectra is selected per epoch to equal the number of spectra in the rarest foreground class.
5. **Data Duplication 1:** Foreground spectra are replicated to equal the number of background spectra, aiming for balance.
6. **Data Duplication 2:** Each class's spectra are duplicated to match the total number of background spectra, ensuring uniform representation across all classes.

Further details about these methods are available in [section 12](#).

The best-performing architecture from the grid search was retrained from scratch using each of these data balancing strategies. Results are detailed in [Table 1](#). An early stopping patience of 500 epochs ensured models reached convergence.

The most effective strategy was **Background Subsampling 1**. Although training without any balancing achieved a nominal accuracy of 97%, this was due to the model learning to classify all spectra as "background," thus failing to provide meaningful predictions. This underscores the necessity of balancing techniques and alternative metrics beyond accuracy to evaluate machine learning models effectively.

The **Class Weighting** approach also performed poorly, with the model again defaulting to predicting "background" for all inputs. This suggests the current architecture lacks the capacity to model background spectra adequately while capturing the characteristics of foreground classes. Increasing the model's complexity, such as adding more convolutional layers, or further reducing the weight assigned to the background class, could improve its performance.

Data Duplication methods showed moderate success, though the model still exhibited a strong bias toward the "background" class. Combining duplication techniques with spectral augmentation could enhance generalization. Among these methods, **Data Duplication 2**, which replicates spectra from each foreground class to match the background, yielded encouraging results. However, duplication-based strategies significantly increased training time, which is a notable drawback.

It is important to note that the architecture was originally optimized using **Background Subsampling 1**, which likely contributes to its strong performance with this method. A comprehensive architecture search across all combinations of data balancing techniques and model configurations would be ideal but is computationally

Table 2: Results of post-processing optimisation for both spatial and spectral metrics.

Model	Optimised for	SC	SCC	SA Factor	F1 Score
1D-ConvNet	Spectral performance	N/A	No Filter	No Filter	0.3653
		N/A	0.1	6.5	0.3502
1D-ConvNet + SAM2	Spatial performance	0.1	No Filter	No Filter	0.5574
		0.4	0.3	6.5	0.5729

prohibitive when using grid search. Future work may explore more sophisticated optimization algorithms to efficiently handle this multidimensional parameter space.

For subsequent sections, the 1D-ConvNet model referenced is the one trained using the **Background Sub-sampling 1** method with the architecture determined through grid search.

5.3 Spatial-spectral Segmentation Algorithms

The spatial-spectral approach integrates spectral classification with spatial segmentation to enhance detection capabilities in HSI. This method involves evaluating four combinations of spectral and spatial models: RF + FastSAM, RF + SAM2, 1D-ConvNet + FastSAM, and 1D-ConvNet + SAM2. Each combination capitalizes on the strengths of its spectral and spatial components to achieve optimal performance.

The spatial-spectral segmentation process operates in two stages. Initially, spectra are classified using a spectral classifier. Subsequently, the RGB image is over-segmented by a spatial segmentation algorithm. The results from these steps are merged, mapping spectral classifications onto the segmentation masks to delineate objects of interest.

Spectral classification relies on machine learning models to analyze HS data directly. The Random Forest RF classifier identifies patterns using an ensemble of decision trees, allowing it to learn relationships within the spectral inputs. In contrast, the 1D-ConvNet extracts hierarchical features using its deep learning architecture, combining convolutional and fully connected layers. Despite differing methodologies, both classifiers are fundamentally data-driven, leveraging the patterns within the HS dataset to make predictions.

For spatial segmentation, two distinct models are used: FastSAM and SAM2. FastSAM is optimized for computational efficiency and speed, making it well-suited for real-time or large-scale applications. Conversely, SAM2 focuses on achieving higher accuracy by employing more resource-intensive spatial segmentation techniques, prioritizing precision over speed.

Segments identified by either FastSAM or SAM2 are classified based on the most frequently occurring non-background class within the segment, determined from the predictions of the N spectra it contains. The class of each individual spectrum within a segment is assigned based on the highest classification probability output by the spectral detector, whether RF or 1D-ConvNet.

This integrated approach leverages the strengths of spectral and spatial models, enabling effective detection and segmentation of objects within HS images.

5.3.1 Detection Post-processing via Confidence Thresholds and Similarity Filtering

To enhance the performance of spectral and spatial-spectral algorithms, we introduce three hyperparameters: Spectral Classification Confidence (SCC), Spectral Angle Factor (SA Factor), and Segmentation Confidence (SC). The first two are applicable to spectral detections, while SC is specific to spatial-spectral algorithms. These hyperparameters facilitate filtering and refining model outputs, improving both detection accuracy and segmentation quality.

The **SC** parameter applies exclusively to spatial-spectral models. It is used to validate whether a segmentation mask represents a true positive detection. It is calculated as:

$$SC = \frac{\sum^N pr_n == c}{N} \quad (1)$$

Here, pr_n denotes the spectral classification of spectrum $n \in N$, c is the predicted class of the segment, and N represents the set of spectra within the segment. By applying a threshold to SC, predictions with low confidence are filtered out. This approach discards segmentation masks covering large background regions or segments with ambiguous spectral signatures, thereby refining the final output.

SCC introduces a threshold-based confidence mechanism for spectral detections. Spectra with a maximum class probability below the threshold are reassigned to the background class. This is described as:

$$label = \begin{cases} \text{argmax}(P) & \text{if } \max(P) \geq \text{thresh} \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

Here, P represents the predicted probability distribution across classes, including background, Royal Gala Apple, Pear, and Cox Apple. This filtering step enhances the specificity of spectral predictions by ensuring low-confidence detections are excluded.

The **SA Factor** applies a similarity-based filter, leveraging SAs to compare individual spectra with class-mean spectra derived from training data. The SA calculation is formalized as:

$$SA(u, v) = \theta = \cos^{-1} \left(\frac{u \cdot v}{\|u\| \|v\|} \right) \quad (3)$$

Using Equation 4, spectra are filtered based on their SA values relative to predefined thresholds for each class:

$$label = \begin{cases} label_{initial} & \text{if } \theta_c \leq \text{thresh}_c \\ \text{argmin}(SA) & \text{else if } \min(SA) \leq \text{thresh}_k \\ 0 & \text{else} \end{cases} \quad (4)$$

The SA threshold for each class is determined by the SA Factor, as described in Equation 5:

$$\text{thresh}_c = \text{SA Factor} \times \text{STD}(SA(S_c, \text{Mean}(S_c))) \quad (5)$$

Here, standard deviation of the SA represents the standard deviation of SAs between spectra within a class and the class mean spectrum in the training data. Consequently, classes with high variability have higher thresholds, whereas uniform classes have stricter thresholds for the same SA Factor.

A grid search was conducted to determine the optimal values for SCC, SA Factor, and SC, optimizing for the F1 score of the spatial-spectral segmentation predictions. This approach prioritizes final segmentation performance over intermediate spectral detections, although optimizing the latter remains a viable strategy.

Preliminary results suggest that appropriate post-processing hyperparameters can substantially enhance performance. For spectral detection, SCC and SA Factor filtering led to improved F1 scores, while SC optimization refined spatial segmentation predictions, resulting in a 1.5% increase in the segmentation F1 score. The optimal parameters for spectral detection differed from those for spatial segmentation due to two key factors: the high false-negative rate of the unfiltered spectral detector and the sparsity of spectral sampling.

The high false-negative rate implies that many true positive segmentation masks contain few positive spectral detections. Simultaneously, sparse spectral sampling means false positives are often spatially isolated, minimizing their impact on segmentation results at suitable SC thresholds. Loosening spectral detection filters to reduce false negatives improves the true positive rate for spatial segmentation.

Finally, a finer-grained grid search or more sophisticated optimization techniques could yield further performance improvements.

5.3.2 RGB Detection and Segmentation with a Fine-tuned YOLO model

The RGB detection and segmentation model utilises a fine-tuned YOLOv8s architecture, originally pre-trained with Ultralytics weights, specifically adapted for segmentation tasks. The model underwent fine-tuning on the orchard dataset to enhance its performance in predicting bounding boxes for objects.

To prepare the dataset for YOLOv8s, the RGB scene views were extracted from hyperspectral images, and segmentation masks were converted into polygon corner coordinates outlining the labeled objects. These data were exported in the YOLOv5 segmentation format, ensuring compatibility with the model.

The RGB scene views and segmentation masks were cropped to the central 1920x1920 region prior to data export. This cropping was necessary for two primary reasons: first, the labeled data included segmentation masks only for this region; second, the Living Optics Camera provides spectral samples exclusively within this area. By standardizing the dataset in this manner, it became easier and fairer to compare the performance of various detection and segmentation models.

The fine-tuning process was conducted using the Ultralytics fine-tuning framework. Key hyperparameters included a maximum of 10,000 training epochs, with an early stopping patience of 200 epochs to prevent over-fitting. The model was trained using 960x960 image dimensions, the Adam optimizer with default settings, and a dropout rate of 40%. Default data augmentation parameters provided by the Ultralytics framework were employed during training by calling the 'YOLO.train()' function.

6. PERFORMANCE COMPARISON

This research employs a comprehensive performance evaluation framework to rigorously analyse the capabilities of various models in HSI and RGB detection and segmentation tasks. By examining both spectral and spatial characteristics, the framework provides an in-depth understanding of each model's strengths and limitations, ensuring a nuanced evaluation of detection efficacy. The metrics used in the analysis facilitate a robust, multidimensional assessment of the models' performance.

Accuracy serves as the baseline measure, providing an overall indication of prediction correctness. However, due to the limitations of accuracy in capturing nuanced performance, particularly in datasets with imbalanced class distributions, additional metrics such as Precision and Recall are considered. Precision evaluates the model's ability to correctly identify positive instances while minimising false positives. Recall, equivalent to the true positive rate, assesses the model's capability to detect all actual positive instances. These metrics, used in tandem, offer a detailed view of detection capabilities.

The F1 score balances Precision and Recall, providing a single measure of performance. The F2 score, which places greater emphasis on Recall, is particularly valuable in scenarios where minimising false negatives is critical. This is especially important in spectral detection tasks, where undetected instances can have a significant impact on performance.

For localisation evaluation, the framework incorporates IoU in two forms: Spectral IoU and Spatial IoU. Spectral IoU assesses the effectiveness of spectral feature detection, particularly in the context of the Living Optics Camera, which uses a sparse spectral sampling mask comprising 4,384 points across the scene. It compares the predicted class area within this mask against the ground truth area. Spatial IoU, on the other hand, evaluates localisation accuracy using dense RGB segmentation masks, offering insights into the geometric accuracy of spatial predictions. To complement IoU, mAP summarises precision across multiple thresholds, providing a holistic measure of detection quality.

Additional evaluation metrics include G-Mean⁴⁹ and the Matthews Correlation Coefficient (MCC).⁵⁰ G-Mean reflects the balance between sensitivity and specificity, ensuring equitable performance across classes. MCC provides a comprehensive measure derived from all elements of the confusion matrix, making it particularly suitable for imbalanced datasets.

Error patterns are further explored through False Positive Rate (FPR) and False Negative Rate (FNR). These metrics, when considered alongside Recall, offer deeper insights into misclassification trends and their underlying causes.

Table 3: Experimental results of spectral classifiers

Metrics	RF	1D-ConvNet	1DConvNet + Similarity Filtering
mAP@50	0.2266	0.3768	0.3254
F1 score	0.1836	0.3653	0.3502
F2 score	0.1637	0.5077	0.4744
Recall	0.1569	0.7840	0.7276
Precision	0.3059	0.2622	0.2649
Spectral_IoU	0.1170	0.2319	0.2192
TNR	0.9949	0.5077	0.4744
FNR	0.8431	0.2160	0.2724
FPR	0.0051	0.0895	0.0863
G-Mean	0.2694	0.8387	0.8018
MCC	0.1936	0.3688	0.3384
Accuracy	0.5229	0.9082	0.9103
GFLOPs	1.05	0.09	0.09

Table 4: Experimental results of spatial-spectral classifiers

Metrics	RF + FastSAM	RF + SAM2	1D-ConvNet + FastSAM	1D-ConvNet + SAM2	1DConvNet + SAM2 + Similarity Filtering
mAP@50	0.4572	0.4238	0.5738	0.8192	0.8210
F1 score	0.3171	0.2978	0.3619	0.6396	0.6416
F2 score	0.2947	0.2820	0.3402	0.6596	0.6620
Recall	0.2893	0.2813	0.3399	0.7004	0.7030
Precision	0.4442	0.3920	0.5170	0.6732	0.6741
Spatial_IoU	0.2531	0.2315	0.2623	0.4866	0.4886
TNR	0.9977	0.9971	0.9938	0.9809	0.9809
FNR	0.7107	0.7187	0.6601	0.2996	0.2970
FPR	0.0023	0.0029	0.0062	0.0191	0.0191
G-Mean	0.3722	0.3622	0.4527	0.8157	0.8175
MCC	0.3321	0.3086	0.3848	0.6508	0.6527
Accuracy	0.5585	0.5446	0.6732	0.9750	0.9751
GFLOPS	7.33	5550.52	6.48	5549.56	5549.56

While real-time application performance is not the primary focus of this study, the estimated Giga Floating Point Operations per Second (GFLOPs) for each algorithm are reported. This metric offers a proxy for computational efficiency and resource requirements, which are essential considerations for model deployment.

For the random forest model, GFLOPs per frame are estimated using the formula in Equation 6:

$$GFLOPs_{RF} \approx \frac{\sum_{n=0}^{n=N} Depth_n \times 96 \times 4384}{10^9} \quad (6)$$

Here, 96 represents the number of features per spectrum, and 4,384 corresponds to the number of spectra per frame. Since Gini impurity splits are precomputed at inference, the primary computational cost of the random forest is determined by the number of trees and their respective depths.

The FLOPs for FastSAM, YOLOv8, SAM2, and the 1D-ConvNet models are calculated using PyTorch's flop counter, ensuring consistency and accuracy in performance estimation.

6.1 Experimental Results

6.1.1 Spectral Results

The spectral performance results presented in Table 3 reveal that the 1D-ConvNet significantly outperforms the RF model across several metrics. The 1D-ConvNet demonstrates notably higher Accuracy, Recall, and Spectral

Table 5: Experimental results of spatial-spectral classifiers

Metrics	1DConvNet + SAM2 + Similarity Filtering	YOLOv8s
mAP@50	0.8210	0.7535
F1 score	0.6416	0.5561
F2 score	0.6620	0.4918
Recall	0.7030	0.4586
Precision	0.6741	0.7532
Spatial_IoU	0.4886	0.4098
TNR	0.9809	0.9970
FNR	0.2970	0.5414
FPR	0.0191	0.0030
G-Mean	0.8175	0.6473
MCC	0.6527	0.5721
Accuracy	0.9751	0.9372
GFLOPS	5549.56	6.28

Table 6: Counting rate

Labels	1DConvNet + SAM2 + Similarity Filtering	YOLOv8s
Royal Gala Apple	0.4981	0.2541
Pear	0.7948	0.4802

IoU, underscoring its ability to identify true positives effectively and capture HS features comprehensively. Conversely, the RF exhibits superior Precision and a lower FPR, indicating its conservative approach and fewer false positives. However, its high FNR and low F2 score highlight substantial limitations in detecting true positives—a critical drawback in applications like fruit yield estimation, where accurate detection is paramount.

Visual comparisons between Figure 5a and Figure 5b highlight the RF’s pronounced FNR, as it fails to detect numerous pear spectra. The 1D-ConvNet, while more adept at detecting pears, exhibits a higher FPR, frequently misclassifying background spectra and mistakenly categorising many pear spectra as belonging to the Cox Apple class. Additionally, by applying post-processing in Figure 5c, less relevant spectral detections were eliminated. This leads to reduced False Positives while increasing Precision value.

In this workflow, individual false-positive spectral detections have a less pronounced impact compared to false-negative detections. Given the sparse nature of spectral sampling and the subsequent use of spatial segmentation on the RGB image, false-positive spectral detections are unlikely to propagate to downstream tasks. In other words, such detections are unlikely to occur with sufficient density to result in false-positive spatial-spectral detections.

The misclassification of numerous pear spectra as Cox Apple by the 1D-ConvNet suggests that excluding the very limited Cox Apple spectra from the training data could improve performance. These spectra have clearly acted as a confounding factor for the Pear class, diminishing the model’s overall accuracy.

Class imbalance, particularly between foreground and background regions, posed a significant challenge during the training of the spectral detector. While the 1D-ConvNet employed custom balancing techniques—including background subsampling and foreground duplication—the RF training process could not accommodate such methods. Instead, the RF relied solely on a loss weighting scheme. Although this approach mitigated some bias towards the background class, it could not fully address the data imbalance. A similar loss weighting scheme applied to the 1D-ConvNet also failed to yield satisfactory results, as the 1D-ConvNet suffered from mode collapse, learning to predict only the “background” class for all spectra. In this specific setup, the RF performed better than the 1D-ConvNet, as it achieved a non-zero True Positive Rate (TPR), unlike the collapsed state of the 1D-ConvNet.

The RF’s high FNR becomes particularly problematic when combined with an RGB segmentation algorithm. Since a confidence threshold on spectral classifications is necessary to determine segment classification, the spec-

tral model's high FNR exacerbates the FNR in the combined system. A single true-positive spectral classification is often insufficient to classify a segment as true positive, further amplifying the FNR when spectral and spatial components are integrated.

6.1.2 Spatial-Spectral Results

The evaluation of the four spatial-spectral segmentors and the spatial-only YOLOv8s model revealed distinct strengths and weaknesses across different approaches. The results, summarised in [Table 4](#) and [Table 5](#), alongside object-specific counting rates in [Table 6](#), highlight the comparative performance of these methods.

Among all configurations, the 1D-ConvNet + SAM2 combination emerged as the top performer across most metrics. While YOLOv8s demonstrated strong Spatial IoU and achieved the highest Precision, reflecting its spatial segmentation strengths, it struggled in metrics such as Recall and F1 score. This deficiency, particularly its high FNR, severely limits its suitability for downstream tasks such as yield estimation, where both counting and sizing of fruit are critical. Over half of all fruit were missed by YOLOv8s.

The object-specific detection outcomes in [Table 6](#) further emphasise the advantages of the 1D-ConvNet + SAM2 configuration. With counting rates of 0.4597 for Royal Gala Apple and 0.7926 for Pear, it outperformed all other configurations, including those based on RF models and YOLOv8s. These results confirm the effectiveness of 1D-ConvNet + SAM2 in integrating spectral and spatial data for precise instance counting. Its superior Recall and F2 score underscore its strength in detecting true positives, a critical factor for tasks requiring high accuracy in instance detection. By contrast, while the RF demonstrated higher Precision, it suffered from substantially lower Recall, resulting in missed detections and reduced counting accuracy. This disparity highlights the importance of recall-focused metrics, such as the F2 score, when evaluating models in scenarios prioritising true positive detection.

The RF + SAM2 combination served as a baseline for spatial-spectral detection. However, its low Recall significantly hindered its utility in downstream tasks due to many missed detections.

In contrast, the 1D-ConvNet + SAM2 combination delivered superior performance by learning spectral features directly from the data. The use of categorical cross-entropy loss and batch normalisation stabilised the training process, while a data subsampling scheme improved dataset balance. These enhancements enabled the 1D-ConvNet to surpass the baseline model in all key metrics. Furthermore, integrating SAM2 for spatial segmentation allowed the model to effectively combine spectral and spatial features, establishing a robust and consistent framework for spatial-spectral detection. This combination significantly outperformed the RF + SAM2 baseline, particularly in classifying spectral data.

The 1D-ConvNet + SAM2 with similarity filtering combination emerged as the most effective spatial-spectral detector, achieving the highest performance across key metrics. It recorded a Spatial IoU of 0.4886 and an mAP@50 of 0.8210, demonstrating its capability to seamlessly integrate spectral and spatial features.

Although YOLOv8s exhibited strong spatial segmentation performance with a Spatial IoU of 0.4098 and Precision of 0.7532, its low Recall and high FNR render it unsuitable for real-world scenarios involving occlusion and shadowing.

By contrast, the 1D-ConvNet + SAM2 combination effectively balanced spectral and spatial detection through deep learning for spectral feature extraction and SAM2's advanced spatial segmentation. Its high Recall represents a promising foundation for future detection workflows. However, the significant computational cost associated with SAM2 remains a notable limitation.

The relatively weak performance of the 1D-ConvNet + FastSAM combination highlights the necessity of fine-tuning the RGB segmentor. Despite sharing the same core RGB segmentation model, YOLOv8s significantly outperformed FastSAM due to fine-tuning. Future work will explore fine-tuning FastSAM to achieve results comparable to the 1D-ConvNet + SAM2 model but with considerably reduced computational requirements. As YOLOv8s surpasses the 1D-ConvNet + SAM2 with similarity filtering model in some metrics, fine-tuning FastSAM could reveal the true benefits of integrating RGB and spectral data—namely, significantly improved detection rates with only a modest increase in computational demands.

The segmentation mask differences between FastSAM ([Figure 6a](#)) and YOLOv8s ([Figure 6e](#)) further highlight this point. Unlike YOLOv8s, which correctly identifies nearly all pears in the scene, FastSAM proposes numerous

incorrect segmentation masks. This disparity likely stems from the fine-tuning applied to YOLOv8s but absent for FastSAM.

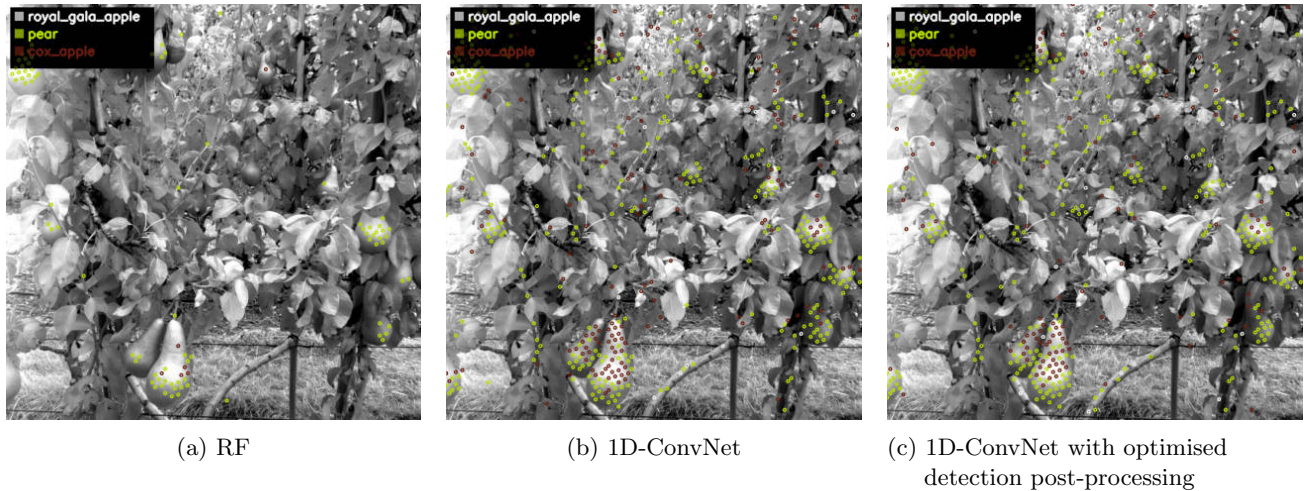


Figure 5: Spectral points of detected masks.

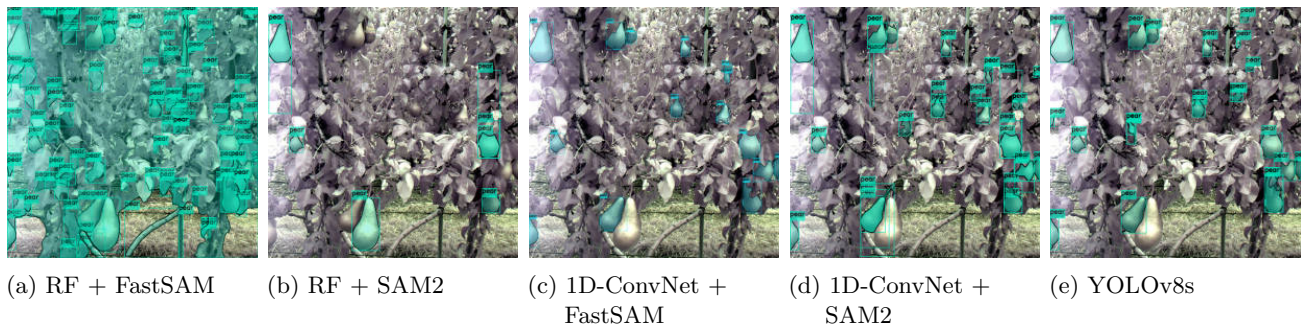


Figure 6: Bounding boxes of detected masks.

7. YIELD ESTIMATION VIA OBJECT TRACKING

With the completion of the spatial-spectral detector, we proceed to develop a tracking algorithm that builds upon the best-performing spatial-spectral detector.

While counting the number of detections per frame is straightforward, determining the number of unique detections across a sequence of frames is more challenging. Objects may not be detected on every frame, and variations in both the scene and the camera's movement make it difficult to identify previously detected objects.

To address this, we employ a tracking algorithm that estimates which detections correspond to unique objects and which are simply repeated detections of the same object(s).

7.1 Method

Each detected fruit is assigned a set of 3D coordinates relative to the camera's position in the first frame. This requires estimating the distance between the camera and each detection, which involves generating a metric depth image.

We utilise DepthAnythingV2 (DAv2)^{51,52} with the 'vitl' encoder and the 'VKitti 2' metric depth checkpoint.^{53,54} Although these weights were designed for urban scenes in the VKitti 2 dataset and are not optimal for orchard environments, fine-tuning DAv2 for this domain is beyond the scope of this study.

We set the maximum output depth to 1, as it is simply a scalar that will be calibrated in a later step. Thus, the initial detections will be in ‘uncalibrated units’—units proportional to meters, with a calibration step to convert them to actual measurements.

After applying DAv2 to the RGB image from the Living Optics Camera, we convert the resulting depth image into a point-cloud in uncalibrated units using Equation 7 for each (X, Y) coordinate.

The segmentation mask for each detected fruit is used to identify the centroid of the object, by calculating the mean X_{ucu} , Y_{ucu} , and the corresponding Z_{ucu} coordinates of the detection’s centroid. Here, X_{ucu} , Y_{ucu} , and Z_{ucu} represent the uncalibrated units along the X, Y, and Z axes of the centroid, respectively.

$$P_{ucua} = Z_{ucu} \left(\frac{P_{pa} - c_a}{f_a} \right) \quad (7)$$

Equation 7 converts pixel coordinates to uncalibrated units, using the intrinsic parameters of the Living Optics Camera. P_{ucu} represents the coordinate along axis a (X or Y), $P_{pa} \in [x_p, y_p]$ is the pixel coordinate, f_a is the focal length in pixels for axis a , and c_a is the principal point along axis aa . We assume that the principal point lies at the centre of the image, i.e., $(2048, 2432)/2 = (1024, 1216)$.

$$f_a = \frac{f_{ma}}{PPx} \quad (8)$$

Equation 8 demonstrates the conversion from the focal length in millimetres f_{ma} to pixels, using the pixel pitch PPx (the size of a pixel) in metres. For the Living Optics Camera, $PPx = 2.76e^{-6}m$, and the focal length in the orchard dataset is 8.5mm for both the X and Y axes.



Figure 7: The average size of the black squares is used to estimate the conversion from uncalibrated units to meters.

To convert from uncalibrated units to meters, we use a reference object of known size, manually labelled in two scenes. We use black squares from a printed chequerboard pattern (shown in Figure 7), each square with a side length of 2.5 cm. By measuring the size of these squares in uncalibrated units, we derive a calibration factor for converting from uncalibrated units to meters.

$$L_m = L_{ucu} * J \quad (9)$$

Equation 9 describes how to convert a length LL in uncalibrated units to meters using the conversion factor J , which is calculated as $J = 0.025 / \text{Mean}(L_{\text{square-ucu}})$. In practice, we found that calibrating the X and Y

axes separately yields better results, likely because the depth image and RGB image have different resolutions along these axes. We set $J_x = 15.524$ and $J_y = 15.863$).

This approach to generating and calibrating a point-cloud has several limitations:

1. DAv2 is computationally demanding.
2. Depth measurements in uncalibrated units vary depending on the resolution of the input image.
3. The requirement for a calibration object reduces the generalisability of this method.
4. The calibration factor JJ may not be valid for scenes where it was not explicitly calculated.
5. The calibration factor JJ may not be valid at depths other than the calibration object's depth, as DAv2 may produce non-linear depth values.
6. Automatic detection of the calibration object is not implemented, so this process requires offline human intervention.

In,⁵¹ the authors use a single value for J based on the maximum depth of the dataset. However, this does not generalise well to real-world data, as shown by the need to derive axis-dependent calibration factors. This implies that DAv2 should ideally provide linearly varying depth values, reducing concerns about points 4 and 5.

Future work will explore ways to enhance the performance of the monocular depth algorithm, focusing on inference speed, depth accuracy, calibration accuracy, and calibration methods.

To unify the coordinate system of all detections, we employ Iterative Closest Point (ICP)⁵⁵ with a 'PointToPlane' transformation⁵⁶ to compute the transformation between $frame_n$ and $frame_{n-1}$. This transformation allows us to map the coordinates of detections in $frame_n$ to the coordinate system of $frame_{n-1}$. The ICP implementation provided by Open3D is used, with the following hyperparameters: $maxcorrespondencedistance = 5000$, $relativefitness = 1e-07$, $relativermse = 1e-07$ and $maxiterations = 50$. Fine-tuning these parameters may improve results in subsequent fruit counting steps.

$$T_{N \rightarrow 0} = \prod_{n=0}^{n=N} T_{n \rightarrow n-1} \quad (10)$$

By keeping track of the cumulative transformation between $frame_n$ and $frame_0$, we can localise all detections relative to the camera's initial position using Equation 10, where $T_{n \rightarrow n-1}$ is the transform between $frame_n$ and $frame_{n-1}$.

We use a homogeneous definition of the coordinates system thus $T_{n \rightarrow n-1}$ is defined as follows:

$$T = \begin{bmatrix} R_{11} & R_{12} & R_{13} & t_x \\ R_{21} & R_{22} & R_{23} & t_y \\ R_{31} & R_{32} & R_{33} & t_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The 4×4 homogeneous transformation matrix includes a 3×3 rotation matrix (R_{ij}) in the top-left, a translation vector (t_x, t_y, t_z) in the top-right, and a bottom row of $[0, 0, 0, 1]$ to maintain homogeneity.

$$c_0 = T_{n \rightarrow 0} \times c_n \quad (11)$$

By applying Equation 11 to the centroid $c_n \in C_n$ of each detection in frame n , we map it to the coordinate system of the first image frame, $c_0 \in C_0$.

In Figure 8, we observe significant drift in the unified coordinate system between frames. This is likely due to improper hyperparameter selection in the ICP algorithm, as mentioned earlier. The multiplication in Equation 10 means any error in the estimated transformation between two frames compounds over time.

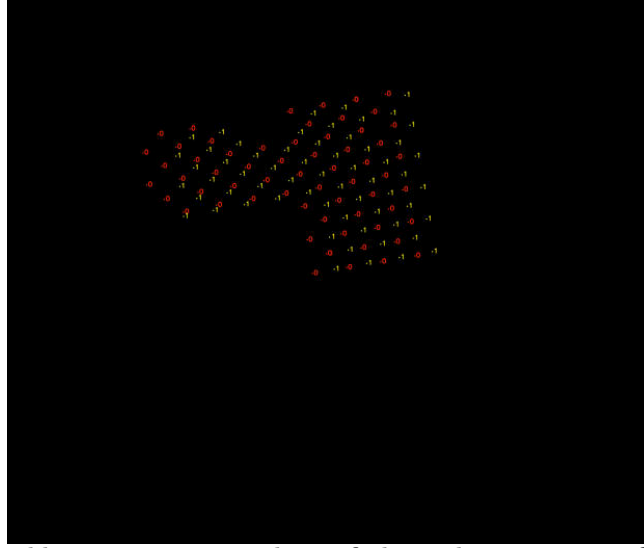


Figure 8: Centroids of the calibration squares in the unified coordinate system. Colour and number represent the frame number in which the squares were imaged.

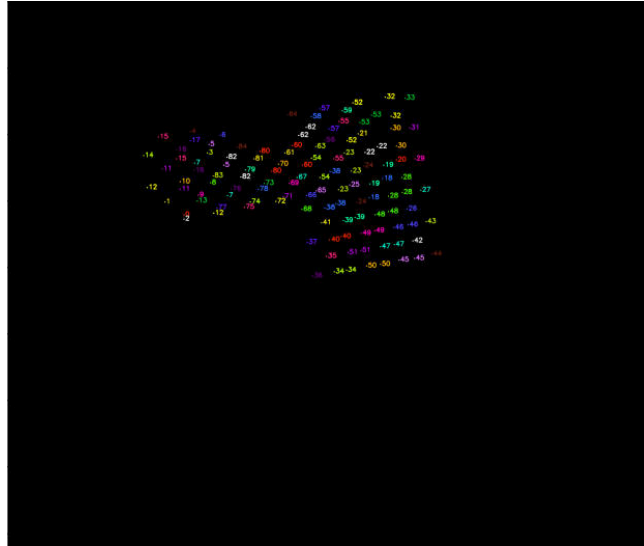


Figure 9: Centroids of the calibration squares in the unified coordinate system. Colour and number represent the estimated unique grouping of the detections.

Once all detections are localised within a single coordinate system, we use agglomerative clustering to group duplicate detections into a linkage matrix and then apply the ‘fcluser’ algorithm to extract the flattened clusters.

We use the Ward variance minimization algorithm [Equation 12](#) to calculate the distance between clusters. The distance between a newly formed cluster u , which consists of clusters s and t , and another cluster v is given by:

$$d(u, v) = \sqrt{\frac{|v| + |s|}{L}d(v, s)^2 + \frac{|v| + |t|}{L}d(v, t)^2 - \frac{|v|}{L}d(s, t)^2} \quad (12)$$

where $L = |v| + |s| + |t|$.

[Figure 9](#) shows how agglomerative clustering groups detections into assumed unique objects. Due to cumulative error in the transformation calculation, the grouping is not perfect.

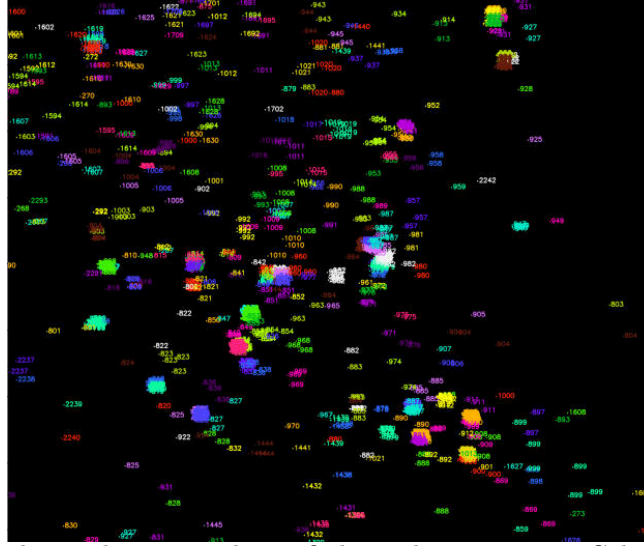


Figure 10: Centroids of the detected pears in the unified coordinate system. Colour and number represent the estimated unique cluster IDs of the detections. Figure is not to scale.

To estimate the total number of fruit on one side of a row of trees in the orchard, we count the number of unique detections for a video of that row.

7.2 Results

Despite the cumulative error in $T_{n \rightarrow 0}$, Figure 10 demonstrates that we have successfully grouped the detections of pears in one row of trees into unique objects.

Table 7: Pear detection statistics by block ID and side of row

Block ID	Side	Total Detections	Total Unique Detections	Estimated Counting Accuracy
7-157	Left	2616	1418	18.91%
7-157	Right	6589	2725	36.33%
7-160	Left	7038	2914	38.85%
7-160	Right	19376	3694	49.25%
Combined	-	35619	10751	35.84%

In Table 7, we do not have a ground truth total count for the unique pears in each of these rows. However, each row is expected to have approximately 15,000 pears. Since we only image one side of the row per video, we assume that we will capture approximately half of the pears. Therefore, we expect to see around 7,500 pears on each side of each row.

8. FRUIT SIZING

8.1 Method

To estimate the size of a fruit, we utilise the segmentation mask provided by the spatial-spectral segmentation algorithm and the point cloud generated by DAv2.

By applying the segmentation mask to the point cloud of a detected fruit, we can estimate its height and width. This is done by selecting the four furthest points within the mask in the left, right, up, and down directions in the imaging plane, and then computing the width as $width = right - left$ and the height as $height = up - down$.

This method does not account for the rotation of the fruit around the Z axis (which is orthogonal to the imaging plane). However, it is computationally simple, and most pears tend to grow vertically due to the increase

in mass in the lower half of the fruit. Apples, being roughly spherical, are not significantly affected by rotational orientation.

8.1.1 Occlusion

A significant challenge in estimating the size of a detected object is occlusion. If an object is partially blocked by an obstacle, such as a leaf, the measured extent of the object will be inaccurate.

To mitigate this effect, we apply an aspect ratio threshold for pears. Since pears are roughly twice as tall as they are wide, any detected pear that does not meet this aspect ratio is likely to be either a false positive or an occluded detection.

We utilise an aspect ratio threshold function, as shown in Equation 13, to filter out detections that are likely to be of occluded pears. Specifically, a detection is considered a partial detection if the height is less than 1.7 times the width of the detection.

$$FullDetection = \begin{cases} True & \text{if } H \geq 1.7W \\ False & \text{else} \end{cases} \tag{13}$$

8.2 Results

Table 8: Pear Size Distribution by Block ID and Side of Row

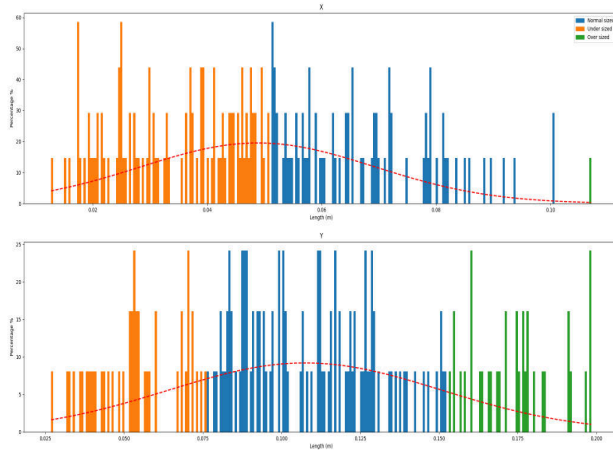
Block ID	Side	Size Distribution (%)			Average Width (cm)	Average Length (cm)
		W < 5.1cm L < 7.6cm	5.1cm ≤ W < 10.2cm 7.6cm ≤ L < 15.2cm	W > 10.2cm L > 15.2cm		
157	Left	56.91%	28.18%	18.78%	4.88 ± 2.04	10.75 ± 4.33
157	Right	44.01%	35.33%	23.55%	5.54 ± 1.99	11.94 ± 3.98
7-160	Left	62.57%	27.19%	12.10%	4.68 ± 1.89	10.09 ± 4.05
7-160	Right	13.11%	54.87%	33.71%	6.88 ± 1.56	14.04 ± 2.96
Combined	-	41.59%	38.08%	22.64%	5.62 ± 2.05	11.88 ± 4.08

Table 8 presents the distribution of pear sizes measured across the sides of the tree rows tested. While the average sizes appear reasonable, the standard deviation of the measurements is quite large. This variance is attributed to both errors in the measurement system—stemming from the calibration method and DAv2 itself—as well as the inclusion of clearly undersized and oversized detections in the calculations.

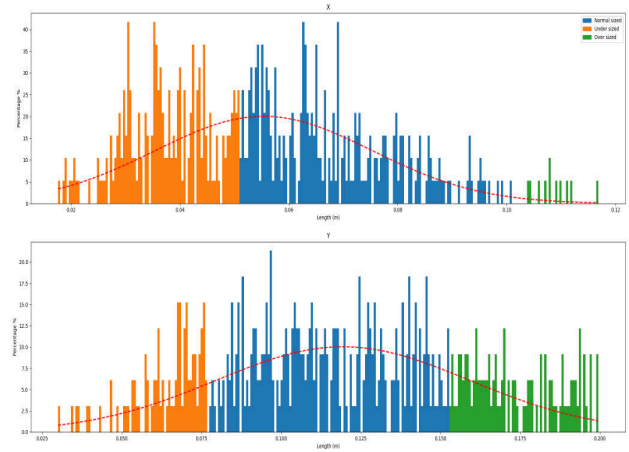
The typical size of a pear is approximately (7.62 cm, 11.43 cm).⁵⁷ To identify pears of the expected size, we apply a 33% threshold on either side of these values.

The largest group of detected pears are undersized. This could indicate an issue with the size estimation method, but it may also result from the improper selection of sizing bands. The expected pear size was sourced from the internet and may not correspond to the variety of pears being imaged. Additionally, the 33% threshold was chosen somewhat arbitrarily.

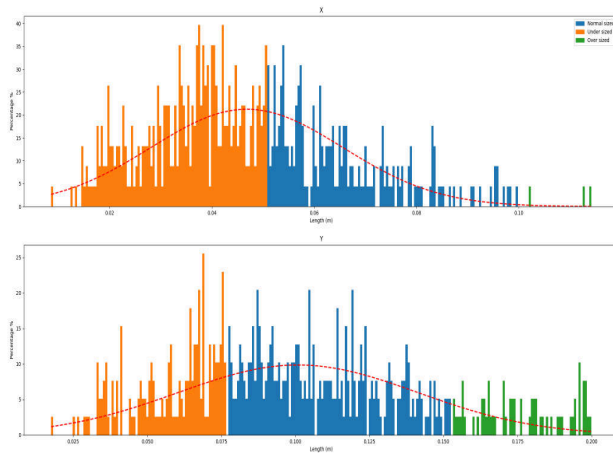
In future work, it will be important to gather ground truth size data for comparison. At a minimum, this should include the mean and standard deviation of pear sizes for each row. We could also consider using size grading thresholds typically applied by farmers, though these are not available at the time of writing.



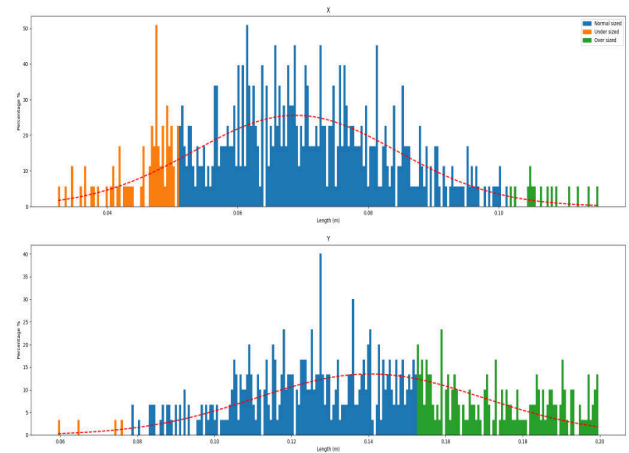
(a) Block 157, left



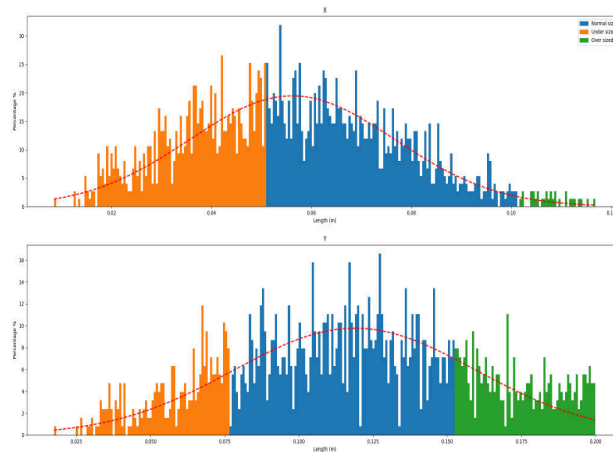
(b) Block 157, right



(c) Block 160, left



(d) Block 160, right



(e) Combined

Figure 11: Distribution of measured pear sizes. The top subplot in each sub-figure (a - e) is for the pear width (X), the bottom for the pear length (Y).

Figure 11 illustrates the distribution of pear sizes measured across each row. For both the width and length of pears, the data from all rows follows a largely normal distribution, which is expected. However, some clear outliers can be observed in the detections, with certain pears appearing too small and others too large. The very small detections are likely to be of occluded pears. The method used to filter out partial detections is rather simplistic, as it only considers the height-to-width ratio and does not take into account the actual size of the detected pears. The unusually large detections might result from errors in the segmentation mask generation, as shown in Figure 6d, where a non-pear object (such as a stick) is mistakenly classified as a pear due to False Positive spectral detections. Implementing minimum object size constraints could help filter out such False Positives, but ultimately, improving the spatial-spectral detection model would provide the best solution.

9. FRUIT RIPENESS ESTIMATION

The accurate estimation of fruit ripeness is essential for the agricultural supply chain, directly influencing harvest timing, post-harvest handling, and consumer satisfaction. Ripeness determines critical fruit attributes such as texture, flavor, sweetness, and shelf life, making it a key quality metric for high-value crops like apples and pears. These attributes also influence market pricing and consumer purchasing decisions. Inconsistent ripeness across a batch can lead to economic losses due to overripe or underripe fruits being rejected during quality inspections or by end-users.

Traditional approaches to ripeness assessment involve destructive and resource-intensive techniques. These include measuring sugar content, represented by total soluble solids (TSS) or Brix, as well as acidity levels such as pH and titratable acidity. These methods typically require juicing the fruit, performing chemical titrations, or using handheld refractometers. While effective, these techniques rely on labor-intensive sampling protocols and require specialized laboratory infrastructure. For large-scale farming operations, the periodic sampling needed to monitor ripeness across orchards over several weeks results in substantial operational costs.

Traditional methods face several challenges that limit their effectiveness in modern agricultural practices. First, their destructive nature results in wasted produce, as sampled fruits cannot be sold or consumed. Moreover, sampling bias often arises because only a small subset of fruits is tested, which may not accurately represent ripeness variability across an orchard. The time-consuming nature of these processes further delays decision-making, which is critical in large-scale operations where timing can impact both yield and quality. Additionally, environmental factors such as temperature and humidity can influence the accuracy of measurements, reducing reproducibility across different conditions. Finally, the dependence on manual intervention and laboratory setups makes these methods incompatible with automation, a key component of modern precision agriculture.

These challenges underline the need for innovative approaches to ripeness estimation that are non-invasive, scalable, and capable of providing real-time insights directly in the field. Hyperspectral imaging (HSI) offers a solution that addresses these limitations by enabling detailed, non-destructive analysis of ripeness indicators.

9.1 Method

Hyperspectral imaging is a powerful technique that captures spectral information across a broad range of wavelengths, allowing for the detection of subtle biochemical and structural changes in the fruit's skin that correlate with ripeness. Unlike traditional methods, HSI provides a non-destructive means of assessment, enabling repeated measurements on the same sample without causing damage. This makes it ideal for longitudinal studies of ripening. Additionally, HSI systems are highly adaptable and can be integrated into automated setups, offering potential for high-throughput analysis in industrial or field environments. Recent advances in HSI technology, such as compact sensors and portable configurations, further enhance its usability for in-field applications.

The spectral data collected through HSI can be processed using statistical and machine learning models to predict ripeness indicators. In this study, partial least squares regression (PLSR) was used to estimate Brix values, a key measure of sweetness and ripeness. Previous studies have shown that HSI combined with PLSR can achieve coefficients of determination (R^2) ranging from 0.2 to 0.7 and root mean square error (RMSE) values between 0.3 and 1.5, depending on the dataset and experimental conditions. This demonstrates the potential of HSI for ripeness estimation while highlighting the need for further refinement of the methodology.

In our experiments, 40 Royal Gala apple samples were imaged using a Living Optics HSI camera in a controlled laboratory setting. The imaging setup included broadband light sources arranged at a $45/0^\circ$ geometry to ensure consistent illumination. A white Tivet sheet was used to facilitate reflectance conversion. To evaluate the robustness of the technique, two types of broadband light sources were employed—halogen and broadband LED. The variation in radiance spectra caused by these lighting conditions is shown in Figure 12b. This setup enabled us to assess the influence of lighting variations on the accuracy of ripeness predictions.

After imaging, the sugar content of the apples was measured using a handheld refractometer, which provides an accuracy of $\pm 0.2^\circ\text{Bx}$. Preprocessing steps included normalization to minimize the effects of surface curvature and smoothing to reduce spectral noise. The spectral data were analyzed in radiance space rather than reflectance space due to challenges in achieving reliable reflectance conversions in the presence of shadows. While this approach requires larger datasets to account for variability, it simplifies preprocessing and makes the model more adaptable to real-world conditions. A linear PLSR model was trained and evaluated using a 50/50 dataset split, correlating spectral features to measured Brix scores.

This decision to work in radiance space, while unconventional, reflects our focus on developing a methodology that can be extended to field applications. Although this approach introduces the need for additional training data, we hypothesize that it offers a more robust pathway for adapting the system to varying environmental conditions. Future work will validate this hypothesis by testing the method under diverse lighting and environmental scenarios.

9.2 Results

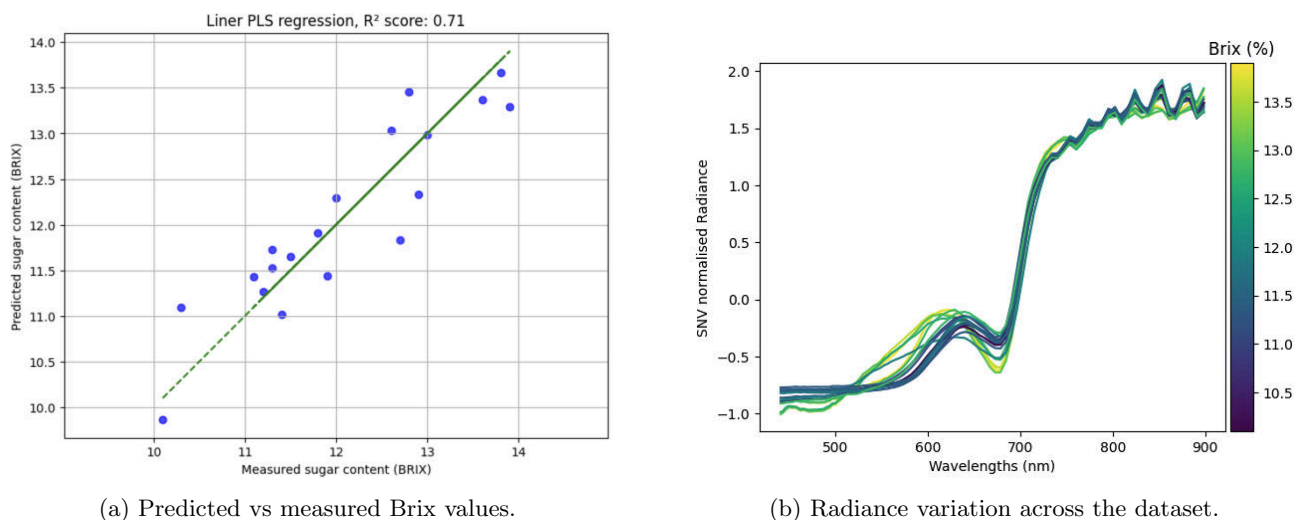


Figure 12: (a) Predicted vs measured values for linear PLSR fitted on the apple Brix dataset. (b) Distribution of radiance spectra across the dataset.

The results, shown in Figure 12a, demonstrate a strong correlation between spectral features and Brix scores, with an R^2 value of 0.71 and an RMSE of 0.4385. These findings confirm that HSI, when combined with a linear PLSR model, can effectively estimate Brix levels. The ability to achieve this level of accuracy using a relatively small dataset highlights the potential of this approach, though further work is required to generalize these findings to diverse conditions.

Despite the controlled nature of the experiments, the study showcases the capability of the Living Optics HSI camera to capture detailed spectral data suitable for ripeness evaluation. Future efforts will focus on expanding the dataset to include a broader range of fruit varieties and environmental conditions. This will enable the development of more robust models and lay the groundwork for field-deployable systems.

While the findings of this study are promising, several areas require further exploration. Expanding the dataset to include a wider variety of apple cultivars and other fruits will improve the generalizability of the

model. Field trials under variable lighting and environmental conditions will be essential to validate the system's robustness and adaptability. Advanced machine learning techniques, such as convolutional neural networks, may further enhance the accuracy of predictions by capturing complex spectral patterns. Additionally, integrating the HSI system with automated sorting and grading equipment could streamline its adoption in industrial settings, enabling large-scale, real-time ripeness assessment.

10. CONCLUSION

We have presented a comprehensive case study on integrating a snapshot hyperspectral, field-portable imaging system to solve critical challenges in orchard fruit quality assessment. Our work introduces a publicly available dataset, benchmarking framework, and system for segmenting, counting, sizing, and estimating fruit ripeness. The dataset, hosted at livingoptics.com/huggingface, is designed to inspire further advancements in hyperspectral computer vision algorithms and foster collaborative innovation in this emerging field.

The results highlight the advantages of combining spectral classification with RGB-based segmentation algorithms. Our proposed 1D-ConvNet + SAM2 with a similarity filter outperformed the RGB-only YOLOv8s approach by nearly 25 percent in Recall, effectively addressing the challenges posed by occlusion, shadowing, and motion blur in real-world agricultural settings. Unlike YOLOv8s, which struggled to detect over half of the fruit due to its reliance on shape and limited color information, the Spatial-Spectral approach achieved over 70 percent detection accuracy. By leveraging hyperspectral data, our method demonstrated the ability to detect partially occluded fruit, requiring only a small visible portion for successful identification.

Beyond fruit detection and segmentation, we developed a two-stage process that uniquely tracks individual fruit across video frames and estimates their size. This approach enables accurate fruit count and size estimation, key metrics for yield and quality assessment in orchards. Additionally, we demonstrated the feasibility of non-destructive ripeness estimation using the Living Optics hyperspectral camera, achieving a Brix correlation with an R^2 of 0.71.

This study underscores the potential of snapshot hyperspectral imaging systems like the Living Optics HSI camera to deliver end-to-end quality assessments of orchard fruit without requiring additional camera systems. By addressing key agricultural challenges, our work paves the way for scalable, efficient, and automated solutions to support modern precision agriculture.

11. FUTURE WORK

This research highlights the potential to further improve spatial-spectral detection, segmentation, counting, and sizing methodologies within hyperspectral imaging systems. Several areas of improvement have been identified that will serve as the basis for future work.

11.1 Spectral detection improvements

Future work will focus on refining the spectral detection algorithm to improve both accuracy and robustness. A comprehensive search for optimal architectures and hyperparameters is essential to achieve the right balance between complexity and performance. Employing Bayesian Optimised Hyperband⁴⁸ will speed up this process by eliminating poorly performing areas in the search space. Exploring alternative loss functions tailored for HS data is also crucial to address imbalanced datasets and improve convergence. Increasing training data through augmentation or more real-world data will help improve generalization and reduce overfitting. Additionally, postprocessing techniques can be introduced to refine detections and minimize False Positives and False Negatives.

11.2 Object segmentation

Currently, segmentation relies on third-party models, and fine-tuning them on domain-specific data is an important next step. However, long-term goals include developing a custom RGB segmentation algorithm optimized for HS tasks, which would emphasize both speed and accuracy. Furthermore, creating a joint spectral and RGB encoder to leverage both modalities directly, instead of treating them separately, could improve performance.

11.3 Monocular depth

Depth estimation currently uses a third-party monocular depth model, and future work will focus on fine-tuning this system to better align with the dataset's characteristics. Developing a custom monocular depth estimation algorithm tailored to HSI applications would offer better control over both accuracy and computational efficiency. Integrating spectral data into the monocular depth algorithm may also enhance performance, as it would provide more detailed feature embeddings. Improving the depth calibration process and automating it will be key to ensuring precise measurements and minimizing calibration errors.

11.4 Tracking

Tracking improvements will focus on unifying the coordinate system used in HS analysis. Optimising ICP hyperparameters will enhance alignment and improve tracking accuracy across multiple frames. For faster processing, replacing ICP with a non-point cloud-based approach like Scale-Invariant Feature Transform combined with homography computation could significantly speed up the process. A key limitation of the current tracking algorithm is its assumption of a static scene and a moving camera; future work should account for object movement within the scene.

11.5 Sizing

The current sizing method relies on bounding box dimensions and can be enhanced by considering object rotation. Future efforts will focus on developing algorithms to estimate the maximum length and width of an object by analyzing its mask, irrespective of orientation. This will provide more accurate size estimates. Capturing ground truth fruit sizes, or at least mean and standard deviation values, would also allow for a more empirical analysis of sizing results.

11.6 Yield estimation

Obtaining ground truth counts of unique fruit would significantly improve the analysis quality in this study. Additionally, combining the counts from the tracking system with size data could enable an estimate of the total volume of fruit. Integrating spectral metrics to assess the health and quality of the fruit could further enhance yield estimation, enabling classifications such as healthy, damaged, or diseased fruit.

11.7 Ripeness Estimation

This study represents an initial test to evaluate HSI's capability in assessing fruit ripeness. While promising, there is a need to expand the dataset in future studies to improve the robustness of predictive models and ensure statistical reliability. Implementing this method in field conditions will be critical for evaluating its real-world applicability, accounting for variables such as ambient lighting and environmental factors. Enhanced predictive algorithms and calibration techniques will also be essential to accommodate diverse fruit varieties and ripening stages, further refining the model's accuracy and generalisability.

12. APPENDIX A

Preliminary experiments with the Random Forest classifier demonstrated the necessity of forming a background class in the training data. This is due to two factors:

1. Random forests must assign a class to all inputs - leading to background spectra being assigned foreground classes if no background class is utilised.
2. The subtle difference between foreground class spectra and background spectra mean that the confidence score of the random forest is not wholly reliable for distinguishing between some foreground and background spectra. Therefore we cannot easily suppress background spectra classified as foreground classes.

The background class is formed of all spectra which do not lie inside a mask from the foreground labelling.

The background class contains many more times the number of spectra that are found in all foreground classes combined - for the training set 1,855,755 background spectra vs 51,285 foreground spectra. This biases both any algorithm trained on the data to preferentially select the background class over all other classes if no attempt is made to either balance the training data or the training loss.

In the case of the random forest, we have no notion of epochs of training or data batching, therefore if we want to make use of all of the training data, we cannot easily remove background spectra to balance the number of foreground vs background spectra.

Instead we have two possible directions to take to achieve a balanced training set:

1. Class weighting
2. Foreground data duplication

12.1 Class weighting

We can provide a set of weights to the random forest training algorithm to reduce the importance of correctly classifying the background class vs the foreground class. These weights are calculated as being inversely proportional to the frequency of the class. This can be done at either the dataset level or at the subsample level and is directly supported in Scikit-learn by passing either 'None', 'balanced' or 'balanced_subsample' to the 'class_weight' argument of the random forest.

We can take a similar approach when training the 1D convolutional neural network, providing a set of class weights which can be used to adjust the loss for each class so as to attempt to avoid biasing the model towards the background class. We only calculate the weighting once, for the entire dataset, rather than on a batch by batch basis as this would require reinitialising the loss function on each batch which would be prohibitively slow during training. Though such an approach might yield better results by re-balancing the loss each batch.

$$w_c = \frac{N}{n_c} \quad (14)$$

In Equation 14, the weighting associated with a class, $w_c \in W$, is inversely proportional to the frequency of class c . N is the total number of spectra in the training data and n_c is the number of spectra of class c in the training data. We then normalise the weights by the maximum, $W_{norm} = \frac{W}{\max(W)}$.

12.2 Foreground data duplication

We could choose to duplicate a random sample of the foreground spectra so that the number of background and foreground spectra match as well as providing balancing across classes.

However, without a data augmentation method, this duplication isn't helpful for training the random forest as duplicated spectra will be classified by the same nodes, without the need for additional nodes to distinguish them. As spectral data augmentation is beyond the scope of this work, we do not explore this approach for the random forest.

For the 1D convolutional classifier, directly duplicating the data is a more reasonable approach, even without data augmentation as this simply biases the loss away from the background class by providing more frequent weight updates calculated from the foreground class losses.

There are two ways in which we can duplicate the foreground data - 1) match the total number of foreground spectra to the total number of background spectra 2) match the number of spectra in each foreground class to the number of spectra in the background class.

$$D_f = \text{duplicate}(D_f, N_b) \quad (15)$$

Equation 15 uses a function called duplicate which will create copies of D_f , the foreground data, so that N_f , the number of foreground spectra matches N_b the number of background spectra.

$$D_{fc} = \text{duplicate}(D_{fc}, N_b) \quad (16)$$

Equation 16 uses the duplicate function to match the number of spectra in class c to the number of background spectra N_b . By applying this method to all $c \in C$, all foreground classes will have the same number of samples as the background class.

12.2.1 Background data sampling

A third option for dataset balancing is available to the 1D convolutional classifier - background data sampling.

As the convolutional classifier is trained for a number of epochs, we can randomly select a sample of the background data at the start of each epoch such that the training set for each consists of N foreground spectra and N background spectra.

Alternatively, we could select the number of background spectra to be equal to the number of spectra in the largest foreground class.

Background data sampling is a good approach to solving the data imbalance problem as it still allows us to leverage all of the background data whilst ensuring that the average loss per epoch is not overly biased towards the background class.

ACKNOWLEDGMENTS

This unnumbered section is used to identify those who have aided the authors in understanding or accomplishing the work presented and to acknowledge sources of funding.

REFERENCES

- [1] Stamford, J., Aciksoz, S. B., and Lawson, T., "Remote sensing techniques: Hyperspectral imaging and data analysis," in [*Photosynthesis : Methods and Protocols*], Covshoff, S., ed., 373–390, Springer US, New York, NY (2024).
- [2] Aref, M. H., Korganbayev, S., Aboughaleb, I. H., Hussein, A. A., Abbass, M. A., Abdlaty, R., Sabry, Y. M., Saccomandi, P., and Youssef, A.-B. M., "Custom hyperspectral imaging system reveals unique spectral signatures of heart, kidney, and liver tissues," *Spectrochimica Acta Part A: Molecular and Biomolecular Spectroscopy* **305**, 123363 (2024).
- [3] Taneepanichskul, N., Hailes, H. C., and Miodownik, M., "Automatic identification and classification of compostable and biodegradable plastics using hyperspectral imaging," *Frontiers in Sustainability* **4**, 1125954 (2023).
- [4] Ma, P., Li, J., Zhuo, Y., Jiao, P., and Chen, G., "Coating condition detection and assessment on the steel girder of a bridge through hyperspectral imaging," *Coatings* **13**(6), 1008 (2023).
- [5] Karim, S., Qadir, A., Farooq, U., Shakir, M., and Laghari, A. A., "Hyperspectral imaging: a review and trends towards medical imaging," *Current Medical Imaging* **19**(5), 417–427 (2023).
- [6] Jaiswal, G., Rani, R., Mangotra, H., and Sharma, A., "Integration of hyperspectral imaging and autoencoders: Benefits, applications, hyperparameter tuning and challenges," *Computer Science Review* **50**, 100584 (2023).
- [7] Guerri, M. F., Distant, C., Spagnolo, P., Bougourzi, F., and Taleb-Ahmed, A., "Deep learning techniques for hyperspectral image analysis in agriculture: A review," *ISPRS Open Journal of Photogrammetry and Remote Sensing*, 100062 (2024).
- [8] Ram, B. G., Oduor, P., Igathinathane, C., Howatt, K., and Sun, X., "A systematic review of hyperspectral imaging in precision agriculture: Analysis of its current state and future prospects," *Computers and Electronics in Agriculture* **222**, 109037 (2024).
- [9] Shuai, L., Li, Z., Chen, Z., Luo, D., and Mu, J., "A research review on deep learning combined with hyperspectral imaging in multiscale agricultural sensing," *Computers and Electronics in Agriculture* **217**, 108577 (2024).

- [10] Chen, Y., Lin, Z., Zhao, X., Wang, G., and Gu, Y., “Deep learning-based classification of hyperspectral data,” *IEEE Journal of Selected topics in applied earth observations and remote sensing* **7**(6), 2094–2107 (2014).
- [11] Borsoi, R. A., Imbiriba, T., and Bermudez, J. C. M., “A data dependent multiscale model for hyperspectral unmixing with spectral variability,” *IEEE Transactions on Image Processing* **29**, 3638–3651 (2020).
- [12] Ma, L., Crawford, M. M., and Tian, J., “Local manifold learning-based k -nearest-neighbor for hyperspectral image classification,” *IEEE Transactions on Geoscience and Remote Sensing* **48**(11), 4099–4109 (2010).
- [13] Melgani, F. and Bruzzone, L., “Classification of hyperspectral remote sensing images with support vector machines,” *IEEE Transactions on geoscience and remote sensing* **42**(8), 1778–1790 (2004).
- [14] Ham, J., Chen, Y., Crawford, M. M., and Ghosh, J., “Investigation of the random forest framework for classification of hyperspectral data,” *IEEE Transactions on Geoscience and Remote Sensing* **43**(3), 492–501 (2005).
- [15] Li, J., Bioucas-Dias, J. M., and Plaza, A., “Semisupervised hyperspectral image segmentation using multinomial logistic regression with active learning,” *IEEE Transactions on Geoscience and Remote Sensing* **48**(11), 4085–4098 (2010).
- [16] Hong, D., Han, Z., Yao, J., Gao, L., Zhang, B., Plaza, A., and Chanussot, J., “Spectralformer: Rethinking hyperspectral image classification with transformers,” *IEEE Transactions on Geoscience and Remote Sensing* **60**, 1–15 (2021).
- [17] Zabalza, J., Ren, J., Zheng, J., Zhao, H., Qing, C., Yang, Z., Du, P., and Marshall, S., “Novel segmented stacked autoencoder for effective dimensionality reduction and feature extraction in hyperspectral imaging,” *Neurocomputing* **185**, 1–10 (2016).
- [18] Lee, H. and Kwon, H., “Going deeper with contextual cnn for hyperspectral image classification,” *IEEE Transactions on Image Processing* **26**(10), 4843–4855 (2017).
- [19] Hsieh, T.-H. and Kiang, J.-F., “Comparison of cnn algorithms on hyperspectral image classification in agricultural lands,” *Sensors* **20**(6), 1734 (2020).
- [20] Hu, W., Huang, Y., Wei, L., Zhang, F., and Li, H., “Deep convolutional neural networks for hyperspectral image classification,” *Journal of Sensors* **2015**(1), 258619 (2015).
- [21] Sun, H., Zheng, X., Lu, X., and Wu, S., “Spectral-spatial attention network for hyperspectral image classification,” *IEEE Transactions on Geoscience and Remote Sensing* **58**(5), 3232–3245 (2019).
- [22] Vaddi, R. and Manoharan, P., “Hyperspectral image classification using cnn with spectral and spatial features integration,” *Infrared Physics & Technology* **107**, 103296 (2020).
- [23] Yu, C., Zhao, M., Song, M., Wang, Y., Li, F., Han, R., and Chang, C.-I., “Hyperspectral image classification method based on cnn architecture embedding with hashing semantic feature,” *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing* **12**(6), 1866–1881 (2019).
- [24] Audebert, N., Le Saux, B., and Lefèvre, S., “Deep learning for classification of hyperspectral data: A comparative review,” *IEEE geoscience and remote sensing magazine* **7**(2), 159–173 (2019).
- [25] Li, S., Song, W., Fang, L., Chen, Y., Ghamisi, P., and Benediktsson, J. A., “Deep learning for hyperspectral image classification: An overview,” *IEEE Transactions on Geoscience and Remote Sensing* **57**(9), 6690–6709 (2019).
- [26] Wang, N.-N., Sun, D.-W., Yang, Y.-C., Pu, H., and Zhu, Z., “Recent advances in the application of hyperspectral imaging for evaluating fruit quality,” *Food analytical methods* **9**, 178–191 (2016).
- [27] Min, D., Zhao, J., Bodner, G., Ali, M., Li, F., Zhang, X., and Rewald, B., “Early decay detection in fruit by hyperspectral imaging—principles and application potential,” *Food Control* **152**, 109830 (2023).
- [28] Rajkumar, P., Wang, N., Elmasry, G., Raghavan, G., and Garipey, Y., “Studies on banana fruit quality and maturity stages using hyperspectral imaging,” *Journal of food engineering* **108**(1), 194–200 (2012).
- [29] Pu, Y.-Y., Feng, Y.-Z., and Sun, D.-W., “Recent progress of hyperspectral imaging on quality and safety inspection of fruits and vegetables: a review,” *Comprehensive Reviews in Food Science and Food Safety* **14**(2), 176–188 (2015).
- [30] Yang, C., Lee, W. S., and Gader, P., “Hyperspectral band selection for detecting different blueberry fruit maturity stages,” *Computers and Electronics in Agriculture* **109**, 23–31 (2014).

- [31] Wei, X., Liu, F., Qiu, Z., Shao, Y., and He, Y., “Ripeness classification of astringent persimmon using hyperspectral imaging technique,” *Food and Bioprocess Technology* **7**, 1371–1380 (2014).
- [32] Khort, D. O., Kuttyrev, A., Smirnov, I., Andriyanov, N., Filippov, R., Chilikin, A., Astashev, M. E., Molkova, E. A., Sarimov, R. M., Matveeva, T. A., et al., “Enhancing sustainable automated fruit sorting: Hyperspectral analysis and machine learning algorithms,” *Sustainability* **16**(22), 10084 (2024).
- [33] Gutiérrez, S., Wendel, A., and Underwood, J., “Ground based hyperspectral imaging for extensive mango yield estimation,” *Computers and Electronics in Agriculture* **157**, 126–135 (2019).
- [34] Syazwani, R. W. N., Asraf, H. M., Amin, M. M. S., and Dalila, K. N., “Automated image identification, detection and fruit counting of top-view pineapple crown using machine learning,” *Alexandria Engineering Journal* **61**(2), 1265–1276 (2022).
- [35] Qureshi, W. S., Payne, A., Walsh, K., Linker, R., Cohen, O., and Dailey, M., “Machine vision for counting fruit on mango tree canopies,” *Precision Agriculture* **18**, 224–244 (2017).
- [36] Vasconez, J. P., Delpiano, J., Vougioukas, S., and Cheein, F. A., “Comparison of convolutional neural networks in fruit detection and counting: A comprehensive evaluation,” *Computers and Electronics in Agriculture* **173**, 105348 (2020).
- [37] Häni, N., Roy, P., and Isler, V., “Apple counting using convolutional neural networks,” in [2018 IEEE/RSJ international conference on intelligent robots and systems (IROS)], 2559–2565, IEEE (2018).
- [38] Maldonado Jr, W. and Barbosa, J. C., “Automatic green fruit counting in orange trees using digital images,” *Computers and Electronics in Agriculture* **127**, 572–581 (2016).
- [39] Miranda, J. C., Gené-Mola, J., Zude-Sasse, M., Tsoulas, N., Escolà, A., Arnó, J., Rosell-Polo, J. R., Sanz-Cortiella, R., Martínez-Casasnovas, J. A., and Gregorio, E., “Fruit sizing using ai: a review of methods and challenges,” *Postharvest Biology and Technology* **206**, 112587 (2023).
- [40] Xuan, G., Gao, C., and Shao, Y., “Spectral and image analysis of hyperspectral data for internal and external quality assessment of peach fruit,” *Spectrochimica acta part A: molecular and biomolecular spectroscopy* **272**, 121016 (2022).
- [41] Shao, H., Li, X., Wang, F., Sun, L., Wang, C., and Hu, Y., “Feasibility study on fruit parameter estimation based on hyperspectral lidar point cloud,” *Journal of Food Measurement and Characterization* **18**(8), 7185–7197 (2024).
- [42] Okamoto, H. and Lee, W. S., “Green citrus detection using hyperspectral imaging,” *Computers and electronics in agriculture* **66**(2), 201–208 (2009).
- [43] Luo, B., Yang, C., Chanussot, J., and Zhang, L., “Crop yield estimation based on unsupervised linear unmixing of multirate hyperspectral imagery,” *IEEE Transactions on Geoscience and Remote Sensing* **51**(1), 162–173 (2012).
- [44] Yang, W., Nigon, T., Hao, Z., Paiao, G. D., Fernández, F. G., Mulla, D., and Yang, C., “Estimation of corn yield based on hyperspectral imagery and convolutional neural network,” *Computers and Electronics in Agriculture* **184**, 106092 (2021).
- [45] Safren, O., Alchanatis, V., Ostrovsky, V., and Levi, O., “Detection of green apples in hyperspectral images of apple-tree foliage using machine vision,” *Transactions of the ASABE* **50**(6), 2303–2313 (2007).
- [46] Chappell, S., Wang, R., Hornett, S., and Spanellis, A., “A VIS-NIR hyperspectral video imager,” in [Photonic Instrumentation Engineering XI], Busse, L. E. and Soskind, Y., eds., **PC12893**, PC1289307, International Society for Optics and Photonics, SPIE (2024).
- [47] Kingma, D. P. and Ba, J., “Adam: A method for stochastic optimization,” *CoRR* **abs/1412.6980** (2014).
- [48] Wang, J., Xu, J., and Wang, X., “Combination of hyperband and bayesian optimization for hyperparameter optimization in deep learning,” (2018).
- [49] Espindola, R. P. and Ebecken, N. F., “On extending f-measure and g-mean metrics to multi-class problems,” *WIT Transactions on Information and Communication Technologies* **35**, 25–34 (2005).
- [50] Chicco, D. and Jurman, G., “The advantages of the matthews correlation coefficient (mcc) over f1 score and accuracy in binary classification evaluation,” *BMC genomics* **21**, 1–13 (2020).
- [51] Yang, L., Kang, B., Huang, Z., Zhao, Z., Xu, X., Feng, J., and Zhao, H., “Depth anything v2,” *arXiv:2406.09414* (2024).

- [52] Yang, L., Kang, B., Huang, Z., Xu, X., Feng, J., and Zhao, H., “Depth anything: Unleashing the power of large-scale unlabeled data,” in [*CVPR*], (2024).
- [53] Gaidon, A., Wang, Q., Cabon, Y., and Vig, E., “Virtual worlds as proxy for multi-object tracking analysis,” in [*Proceedings of the IEEE conference on Computer Vision and Pattern Recognition*], 4340–4349 (2016).
- [54] Cabon, Y., Murray, N., and Humenberger, M., “Virtual kitti 2,” (2020).
- [55] Besl, P. J. and McKay, N. D., “Method for registration of 3-d shapes,” in [*Sensor fusion IV: control paradigms and data structures*], **1611**, 586–606, Spie (1992).
- [56] Chen, Y. and Medioni, G., “Object modelling by registration of multiple range images,” *Image and vision computing* **10**(3), 145–155 (1992).
- [57] Team, S. G., “How big is a pear?,” (2024). Accessed: 2024-12-06.